SDS188 A1S2 2026

COMPLETE OFFLINE PRINT PACK

Chapters 6-8 + Chapter 9 Slides 1-51 only

PURPOSE
A paper-first version of the signed-off SDS188 Master Learning System. Use it to learn, retrieve, practise, check answers and rehearse Excel procedures without needing to open the HTML.

Assessment: Saturday 29 August 2026 | Closed book | Invigilated | EMSLearn + Excel competency

Built from the exact signed-off FINAL HTML (SHA-256 cbeff839...b6c4) plus the latest lecturer scope update.

START HERE - CURRENT A1S2 SCOPE

LATEST CHAPTER 9 RULE

For A1S2, Chapter 9 is assessed only up to and including Slide 51. This pack therefore includes full Chapter 9 learning only through Lesson 9.7. Later detailed one-tail worked procedures, one-proportion hypothesis tests, later power calculations, planning/practical-significance and later ethics material are intentionally excluded from the exam-study body.

Chapter	A1S2 scope in this pack	How to use it
6	Continuous distributions - full current chapter learning	Learn routes, then work the lesson checks/Your Turn items.
7	Sampling distributions - full current chapter learning	Prioritise X vs Xbar, SE, known sigma Z, unknown sigma t, CLT, p and two-mean routing.
8	Confidence intervals - C8.01-C8.68 core only	C8.69-C8.75 auditing/population-total material remains HOLD and is not treated as A1S2 core.
9	Slides 1-51 only - Lessons 9.1-9.7	STOP after 9.7 for A1S2 Chapter 9 study. Basic Type I/II, alpha/beta/power meaning remains in scope; later detailed power calculations do not.

ASSESSMENT RULE WARNING

The Chapter 8 AF is a live unsupervised assessment with source/communication restrictions. Use this pack for preparation. Do not assume an AI-generated pack is an allowed source during the live AF. Follow the EMSLearn rules exactly. The actual A1S2 on 29 August is closed book.

A1S2 PRIORITY MAP - 24 TO 29 AUGUST

1 - Chapter 8 AF preparation and completion: Deadline: Friday 28 Aug, 17:30. One attempt, 60 minutes. Revise Ch8 method selection before opening the live AF and complete it independently.

2 - Narrowed Chapter 9 Slides 1-51: Master Lessons 9.1-9.7: H0/H1, Type I/II, alpha/beta/basic power meaning, tails, known-sigma Z, p-values, CI cross-check, unknown-sigma t.

3 - Official A1S2 Practice Test: Deadline: Friday 28 Aug, 23:59. Use it as the closest EMSLearn/Excel format rehearsal. Password in the announcement: maties.

4 - Repair Excel execution errors: Rehearse the exact formula syntax on paper: COUNT, AVERAGE, STDEV.S, SQRT, ROUNDUP, NORM.*, T.* and bootstrap ordering logic.

5 - Chapter 6 revision AF: Ungraded, unlimited attempts. Use a cold attempt as a diagnostic, then repair only what you miss.

6 - Chapter 7 revision AF: Same diagnostic strategy: attempt cold -> identify error -> repair -> reattempt.

7 - Mixed closed-book retrieval: Use the lesson checks and Your Turn pages in this pack, manually respecting the Chapter 9 Slide-51 cutoff.

HOW TO STUDY FROM THIS PACK WITHOUT A COMPUTER

1. Cover the explanation. Read only the lesson title + recognition cues and say aloud which method you would use.
2. Read the explanation and formula/decision object. Then close the page with a sheet of paper and reconstruct it from memory.
3. Rehearse the Excel route by writing the function exactly as you would type it. You are learning the decision + syntax even when no computer is open.
4. Work the Worked Example line by line, then cover it and reproduce the route independently.
5. Answer the Knowledge Check and Your Turn item without looking at the answer key.
6. Use the Answer Key at the back. If wrong, label the error: recognition / object / distribution / tail / formula / arithmetic / interpretation / Excel syntax.
7. Return to the nearest competing method and say the contrast aloud. Then retry the question later.

FAST RULE
Do not reread whole chapters after a mistake. Repair the exact method choice or calculation step that failed.

CONTENTS - PRINT NAVIGATION

Official 2026 formula sheet
Master method-selection maps + Excel function quick reference
Chapter 6 - 10 full Guided Learn lessons
Chapter 7 - 13 full Guided Learn lessons
Chapter 8 - 13 full Guided Learn lessons (C8.69-C8.75 HOLD excluded)
Official Student-t table
Chapter 9 - 7 in-scope Guided Learn lessons (Slides 1-51 only)
Consolidated answer key for lesson checks and Your Turn items
Final traps + pre-assessment checklist + scratch pages

OFFICIAL 2026 A1S2 FORMULA SHEET

This is the supplied 2026 one-page formula sheet. Learn what it gives you - and what it does not give you.

FORMULA SHEET / *FORMULEBLAD*

Probability distribution formulae / *Waarskynlikheidsverdelingsformules*

Distribution / *Verdeling*	**Expected value / *Verwagte waarde***	**Variance / *Variansie***
$f(x) = \lambda e^{-\lambda x}, \quad x > 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$

Sample sizes / *Steekproefgroottes*

$$n \geq \frac{z_{\alpha/2}^2 \sigma^2}{e^2},$$

$$n \geq \frac{z_{\alpha/2}^2 \pi(1-\pi)}{e^2},$$

FPC / EPK: $$n \geq \frac{n_0 N}{n_0 + (N-1)}.$$

Inference formulae / *Inferensie formules*

Under certain conditions: / *Onder sekere voorwaardes:*

Parameter	**Statistic / *Statistiek***	**Standard error / *Standaardfout***	**Distribution / *Verdeling***
μ	$\bar{X}$	$\frac{\sigma}{\sqrt{n}}$	$Z \sim N(0,1)$, σ known / *bekend*
μ	$\bar{X}$	$\frac{s}{\sqrt{n}}$	$T \sim t_{n-1}$, σ unknown / *onbekend*
π	p	$\sqrt{\frac{\pi(1-\pi)}{n}}$	$Z \sim N(0,1)$
$\mu_1 - \mu_2$	$\bar{X}_1 - \bar{X}_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$Z \sim N(0,1)$, σ_1, σ_2 known / *bekend*

WHAT YOU STILL HAVE TO KNOW

The sheet does not choose the random object, translate the event/tail, decide CI versus test, choose critical-value versus p-value method, determine t degrees of freedom, apply the bootstrap algorithm, or write the contextual conclusion. Those decisions are trained in the lesson pages that follow.

MASTER METHOD-SELECTION MAP

Cue / object	Choose this route	Execution lock
Ch6 Normal X	One continuous normal observation X	Z=(X-mu)/sigma; match left/right/between/inverse region.
Ch6 Uniform	Constant density on [a,b]	Probability = interval length / total length.
Ch6 Exponential	Waiting time between Poisson events	Align rate/time units; EXPON.DIST for cumulative-left; complement for right tail.
Ch6 Binomial -> Normal	Binomial count with approximation conditions satisfied	Continuity-correct the count boundary by 0.5 before standardising.
Ch7 Xbar known sigma	Sample mean; population sigma known	SE=sigma/sqrt(n); Z/normal sampling route.
Ch7 Xbar unknown sigma	Sample mean; population sigma unknown; sample s	SE=s/sqrt(n); Student t; df=n-1.
Ch7 p	Sample proportion	Centre pi; proportion SE; check approximation conditions.
Ch7 two means	Difference of independent sample means	Freeze order 1-2; centre mu1-mu2; combine variances.
Ch8 mean CI known sigma	Estimate mu, population sigma known	Xbar +/- z* sigma/sqrt(n).
Ch8 mean CI unknown sigma	Estimate mu, sigma unknown	Xbar +/- t* s/sqrt(n), df=n-1.
Ch8 proportion CI	Estimate pi	p +/- z*sqrt[p(1-p)/n].
Ch8 required n	Plan sample size	Choose mean/proportion formula; always ROUND UP; finite N may need n0 adjustment.
Ch8 bootstrap	Parametric CI inappropriate / bootstrap requested	Observed sample -> resample WITH replacement, same n -> statistic -> repeat -> order -> cut tails.
Ch9 mean test known sigma	H0 about mu; sigma known	Set H0/H1 + tail; Z statistic; critical-value or p-value decision.
Ch9 mean test unknown sigma	H0 about mu; sigma unknown	Student t, df=n-1; normality support; critical/p-value route.

EXCEL FUNCTION QUICK REFERENCE - REHEARSE ON PAPER

Function	What it does / when to use it
=COUNT(range)	Count numeric observations; often used to obtain n from raw data.
=AVERAGE(range)	Calculate Xbar from raw data.
=STDEV.S(range)	Calculate sample s when population sigma is unknown.
=SQRT(value)	Square root; used throughout SE formulas.
=ROUNDUP(value,0)	Required sample size: always round upward.
=NORM.S.DIST(z,TRUE)	Standard normal cumulative-left probability Phi(z).
=NORM.S.INV(p)	Standard normal z boundary for cumulative-left probability p.
=NORM.DIST(x,mu,sd,TRUE)	Normal cumulative-left probability in original units.
=NORM.INV(p,mu,sd)	Normal boundary in original units.

Function	What it does / when to use it
=T.DIST(t,df,TRUE)	Student-t cumulative-left probability.
=T.DIST.2T(ABS(t),df)	Two-tail t p-value.
=T.INV(p,df)	Student-t boundary for a cumulative-left probability.
=T.INV.2T(alpha,df)	Positive two-tail t critical magnitude for total two-tail probability alpha.
=EXPON.DIST(x,lambda,TRUE)	Exponential cumulative-left probability P(X<=x); align units first.
=SMALL(range,k) / =LARGE(range,k)	Select ordered bootstrap statistics when the course/question gives the cutoff position.

CHAPTER 6

Continuous Distributions

SCOPE
Full current Chapter 6 learning layer.

Offline tracker: [ ] 6.1 [ ] 6.2 [ ] 6.3 [ ] 6.4 [ ] 6.5 [ ] 6.6 [ ] 6.7 [ ] 6.8 [ ] 6.9 [ ] 6.10

CHAPTER 6 | LESSON 6.1 | OFFLINE GUIDED LEARN

6.1 - Continuous Probability Foundations

OBJECTIVE
Distinguish discrete from continuous random variables and explain why probability for a continuous variable is area, not the height of the density curve.

Explanation

A discrete variable jumps between countable outcomes. A continuous variable can take any real value in an interval, so measurements such as time, length, height and volume are continuous.

For a continuous distribution the vertical axis is density $f(x)$, not probability. Probability is the area under the density curve over an interval. The full area is 1.

A single point has zero width, so it has zero area. Therefore $P(X=c)=0$. That is also why $P(X<c)=P(X\leq c)$ for continuous X.

SAY ALOUD
Discrete means countable points. Continuous means any measured value in a range. For continuous variables, probability is area under the density curve, so one exact point has probability zero.

Worked example - Why < and ≤ give the same continuous probability

1. Suppose we want $P(X\leq 5)$.
2. Split it as $P(X<5)+P(X=5)$.
3. For continuous X, $P(X=5)=0$.
4. Therefore $P(X\leq 5)=P(X<5)$.
Result: Including or excluding one boundary point does not change a continuous probability.
Verify: If your reasoning depends on a positive probability at exactly one continuous value, stop and repair that idea.

KNOWLEDGE CHECK - answer before looking at the key
Which statement is correct for a continuous random variable X?

A. $P(X=4)=0$
B. $P(X=4)=f(4)$
C. $P(X=4)$ must be positive

YOUR TURN - work independently
Which event has the same probability as $P(X<8)$ when X is continuous?

A. $P(X\leq 8)$
B. $P(X=8)$
C. $P(X>8)$

EXIT CRITERION
Move on when you can explain, without notes, why density height is not probability and why $P(X=c)=0$.

Sources: 2026 Chapter 6 slides §§ continuous foundations | Chapter 6 lecturer transcript: discrete vs continuous / probability as area | Semester 1 Final Boss hub used only as a checked explanation/layout donor

WORKSPACE / RETRIEVAL NOTES

CHAPTER 6 | LESSON 6.2 | OFFLINE GUIDED LEARN

6.2 - Normal Distribution

OBJECTIVE
Read normal notation, predict how μ and σ change the curve, recognise the standard normal distribution, and standardise X to Z.

Explanation

The normal distribution is continuous, symmetric and bell-shaped. Its centre is μ. In a perfectly normal model, mean = median = mode.

In $X \sim N(\mu,\sigma^2)$, the second parameter is variance. If you are given σ^2, take its square root before using the z-score formula.

Changing μ shifts the curve left or right without changing its basic shape. Increasing σ spreads the same total area over a wider range, so the curve becomes wider and flatter.

Any normal X can be translated to the standard normal $Z \sim N(0,1)$ with $z=(x-\mu)/\sigma$. A z-score tells you how many standard deviations x lies above or below μ.

SAY ALOUD
Mu sets the centre. Sigma sets the spread. Standardising subtracts the mean and divides by the standard deviation, so z tells me how many standard deviations the value is from the mean.

Worked example - Standardise a value

1. Given $X \sim N(100,50^2)$ and $x=200$.
2. Identify $\mu=100$ and $\sigma=50$.
3. Compute $z=(200-100)/50$.
4. $z=2$.
Result: $x=200$ is 2 standard deviations above the mean.
Verify: A value above μ must produce a positive z-score; below μ must produce a negative one.

KNOWLEDGE CHECK - answer before looking at the key

If $X \sim N(18,25)$, what standard deviation should you use?

A. 5
B. 25
C. $\sqrt{18}$

YOUR TURN - work independently

If $X \sim N(50,4^2)$, what is z when $x=42$?

Answer: _____

EXIT CRITERION
Move on when you can look at $N(\mu,\sigma^2)$, identify μ and σ correctly, predict the effect of changing each parameter, and standardise one value without help.

Sources: 2026 Chapter 6 slides pp. 1-2 | 2023/departmental Chapter 6 slides: normal shape and standardisation | Chapter 6 lecturer transcript

WORKSPACE / RETRIEVAL NOTES

CHAPTER 6 | LESSON 6.3 | OFFLINE GUIDED LEARN

6.3 - Normal Regions & Forward Probability

OBJECTIVE
Translate "below", "above" and "between" into the correct shaded region, standardise boundaries, and calculate the required normal area.

Explanation

Do not begin with calculator buttons. First draw or identify the region in X-units. The wording determines the region.

Below / less than means the left cumulative area. Above / greater than means a right tail, so use 1-left area. Between means upper left-area minus lower left-area.

Then standardise any X boundary using $z=(x-\mu)/\sigma$. The course Excel routes are NORM.DIST(x,μ,σ,TRUE) for X directly and NORM.S.DIST(z,TRUE) for Z.

SAY ALOUD
Below is left. Above is one minus left. Between is upper left area minus lower left area. I draw the region first, then standardise, then calculate.

Worked example - Official download-time example: P(X<18.6)

1. Given $X\sim N(18,5^2)$. We want the area left of 18.6.
2. Standardise: $z=(18.6-18)/5=0.12$.
3. Find $\Phi(0.12)=0.5478$.
4. Interpret: about 54.78% of download times are below 18.6 seconds.
Result: $P(X<18.6)=0.5478$.
Verify: 18.6 is slightly above the mean, so a left-tail probability slightly above 0.5 is sensible.

KNOWLEDGE CHECK - answer before looking at the key
Which expression gives a right-tail standard-normal probability $P(Z>z)$?

A. $1-\Phi(z)$
B. $\Phi(z)$
C. $\Phi(z)-1$

YOUR TURN - work independently
For the same $X\sim N(18,5^2)$, find $P(X>18.6)$. Enter a decimal to 4 d.p.
Answer: _____

EXIT CRITERION
Move on when you can identify left/right/between before touching the calculator and can explain why the numerical answer matches the shaded region.

Sources: 2026 Chapter 6 notes §1.3 | Chapter 6 lecturer transcript download-time example | Tutorial 6 + solutions Q1-Q5 | Chapter 6 Excel shortcuts

WORKSPACE / RETRIEVAL NOTES

CHAPTER 6 | LESSON 6.4 | OFFLINE GUIDED LEARN

6.4 - Inverse Normal & Percentiles

OBJECTIVE
Recognise when probability is known but the boundary is unknown, convert wording to cumulative-left area, find z, and transform back to X.

Explanation

Forward normal problems give x and ask for probability. Inverse normal problems give probability or percentile and ask for x.

The inverse tool expects cumulative area to the left. If the wording gives a right-tail percentage, convert it first. Example: top 10% means 10% to the right, so 90% to the left.

Find z from the cumulative-left probability, then return to original units with $x=\mu+z\sigma$. Excel can do this directly with NORM.INV(prob,μ,σ), or use NORM.S.INV(prob) then back-transform.

SAY ALOUD
Inverse normal means the area is known and the boundary is unknown. I convert everything to left area, find z, then use x equals mu plus z sigma.

Worked example - 20th percentile of $N(18,5^2)$

1. We need x with $P(X<x)=0.20$.
2. Inverse standard normal gives $z\approx-0.8416$.
3. Back-transform: $x=18+(-0.8416)(5)$.
4. $x\approx13.792$ seconds.
Result: The 20th percentile is about 13.8 seconds.
Verify: A 20th percentile must lie below the mean, so a negative z and $x<18$ are sensible.

KNOWLEDGE CHECK - answer before looking at the key
The top 10% starts above x. What cumulative-left probability must go into the inverse function?

A. 0.90
B. 0.10
C. 0.05

YOUR TURN - work independently
Tutorial-style: $X\sim N(73,8^2)$. Find the boundary exceeded by only 5% of values. Enter the boundary.
Answer: _____

EXIT CRITERION
Move on when you can translate percentile/top/bottom wording into a left cumulative probability and perform the back-transform without being prompted.

Sources: 2026 Chapter 6 notes §1.3.2 | Percentiles from normal distribution worked sheet | Tutorial 6 + solutions Q2-Q5 | Chapter 6 Excel shortcuts

WORKSPACE / RETRIEVAL NOTES

CHAPTER 6 | LESSON 6.5 | OFFLINE GUIDED LEARN

6.5 - Evaluation of Normality

OBJECTIVE
Judge whether data are approximately normal by integrating descriptive, empirical-rule and graphical evidence instead of relying on one indicator.

Explanation

Real data are rarely perfectly normal. The task is usually to decide whether a normal approximation is reasonable.

Numerical evidence includes mean, median and mode for symmetry and the empirical 68-95-99.7 rule. Mean greater than median suggests right skew; mean less than median suggests left skew.

Graphical evidence includes histograms, frequency polygons, stem-and-leaf displays, boxplots and the normal probability plot. A normal probability plot should be approximately linear.

The strongest conclusion uses several indicators together. A bell-shaped histogram alone does not prove normality, and mean≈median alone does not prove normality.

SAY ALOUD
Normality is a collective judgement. I check centre and symmetry, empirical-rule proportions, and plots. No single check proves the model.

Worked example - Small-sample empirical-rule evidence

1. A worked example has n=15.
2. Observed within 1s: 73% versus theoretical 68%.
3. Observed within 2s: 100% versus theoretical 95%.
4. Observed within 3s: 100% versus theoretical 99.7%.
5. For a small sample, these are treated as reasonably close; combine this with symmetry and plot evidence before concluding.
Result: The example is classified as approximately normal based on the collective evidence.
Verify: Do not write "normal because 73%≈68%". Your conclusion should explicitly combine the available evidence.

KNOWLEDGE CHECK - answer before looking at the key

Which is the strongest conclusion method?

A. Use several numerical and graphical indicators together
B. Mean≈median proves normality
C. A bell-shaped histogram alone proves normality

YOUR TURN - work independently

A dataset has mean≈median, but its histogram is strongly right-skewed. What is the best conclusion?

A. Do not call it normal; the evidence conflicts
B. It is normal because mean≈median
C. Ignore the histogram

EXIT CRITERION
Move on when you can list at least three different normality checks and explain why the conclusion must use them collectively.

Sources: 2026 Chapter 6 notes §§1.4, pp. 8-13 | Chapter 6 lecturer transcript normality discussion | Tutorial 6 Q18-Q20

WORKSPACE / RETRIEVAL NOTES

CHAPTER 6 | LESSON 6.6 | OFFLINE GUIDED LEARN

6.6 - Excel Normality Laboratory

OBJECTIVE
Construct a normal probability plot correctly in Excel and combine it with descriptive/histogram evidence to reach a statistical conclusion.

Explanation

The normal probability plot is built from ordered observed values and the theoretical standard-normal quantiles they should correspond to if the data were normal.

The course workflow is: sort X ascending → create index j → calculate $\alpha_j = j/(n+1)$ → convert α_j to Z with NORM.S.INV → scatterplot Z on the x-axis and ordered X on the y-axis → judge linearity.

This is a separate evidence channel from statistical understanding. Knowing the Excel clicks is not the same as knowing why a straight line supports normality.

SAY ALOUD
Sort X, number the rows j, calculate j over n plus one, turn each probability into Z with NORM.S.INV, plot Z horizontally and X vertically, then judge whether the points are roughly linear.

Worked example - Supplied commuting-time workbook (n=15)

1. For j=1: $\alpha = 1/16 = 0.0625$.
2. NORM.S.INV(0.0625)≈-1.5341.
3. The first ordered X is 24.6. Plot the point (Z,X)=(-1.5341,24.6).
4. Repeat through j=15, ending at $\alpha = 15/16 = 0.9375$ and Z≈1.5341.
5. Use a scatterplot and assess whether the cloud follows an approximately straight line.
Result: The plot is a theoretical-quantile check: approximate linearity supports normality.
Verify: Axes matter: Z belongs on x, ordered X on y in the official workflow.

KNOWLEDGE CHECK - answer before looking at the key
Which plotting position does the current Chapter 6 workflow use?

A. j/(n+1)
B. j/n
C. (j−1)/n

YOUR TURN - work independently
After sorting X and creating j, what Excel formula should convert α in D3 to its theoretical z-value?

A. =NORM.S.INV(D3)
B. =NORM.S.DIST(D3,TRUE)
C. =NORM.INV(D3,0,1)

EXIT CRITERION
Move on when you can reconstruct the six-step normal-probability-plot workflow from memory and explain what linearity means.

Sources: 2026 Chapter 6 notes §1.4 step 3 | Supplied Chapter 6 Excel examples (Solutions).xlsx · Normal plot sheet | T06-data.xlsx · Steel / Utility / Wait | Chapter 6 lecturer transcript normal-plot workflow

WORKSPACE / RETRIEVAL NOTES

CHAPTER 6 | LESSON 6.7 | OFFLINE GUIDED LEARN

6.7 - Uniform Distribution

OBJECTIVE
Recognise a uniform model, identify bounds a and b, use rectangle area for probabilities, and calculate the mean and spread.

Explanation

A uniform distribution is continuous and rectangular. Every equal-width interval inside [a,b] has the same probability because the density height is constant.

The height is $f(x)=1/(b-a)$. The height itself is not a probability; probability is rectangle area = width×height.

For $c<d$ inside [a,b], $P(c<X<d)=(d-c)/(b-a)$. The mean is $(a+b)/2$ and the standard deviation is $\sqrt{(b-a)^2/12}$.

SAY ALOUD
Uniform means a rectangle from a to b. Height is one over b minus a, but probability is area. Interval probability is interval width divided by total width.

Worked example - Tutorial 6 Q8: Uniform(0,10), P(5<X<7)

1. Identify a=0, b=10.
2. Total width is 10. Target interval width is 7-5=2.
3. Probability=2/10=0.20.
4. Equivalent area method: height=1/10=0.1, area=2×0.1=0.20.
Result: P(5<X<7)=0.20.
Verify: The interval uses 20% of the total horizontal width, so 0.20 is visually sensible.

KNOWLEDGE CHECK - answer before looking at the key

For Uniform(a,b), what does 1/(b−a) represent?

A. Density height
B. Every interval probability
C. The mean

YOUR TURN - work independently

For X~Uniform(0,10), find P(2<X<3).

Answer: _____

EXIT CRITERION
Move on when you can draw the rectangle, label a and b, explain why height is not probability, and solve an interval by area without memorising a special case.

Sources: 2026 Chapter 6 notes §§2.1-2.1.5 | Uniform probabilities worked sheet | Tutorial 6 + solutions Q8-Q9 | Chapter 6 lecturer transcript uniform explanation

WORKSPACE / RETRIEVAL NOTES

CHAPTER 6 | LESSON 6.8 | OFFLINE GUIDED LEARN

6.8 - Exponential Distribution

OBJECTIVE
Recognise exponential waiting-time problems, interpret λ as a rate, align units, and solve left-, right- and between-probability questions.

Explanation

The exponential distribution models continuous waiting time between events. It is right-skewed and starts at 0.

λ is a rate per unit time or opportunity, not the waiting time itself. Mean waiting time is $1/\lambda$ and variance is $1/\lambda^2$.

Use P(X≤x)=1-e^(-λx) for a left tail and P(X≥x)=e^(-λx) for a right tail. For between events subtract two cumulative left areas.

Poisson and exponential share the same rate context: Poisson counts how many events occur in a fixed interval; exponential measures the time until/between events.

Before calculating, λ and x must use the same time unit.

SAY ALOUD
Poisson counts events. Exponential waits for the next event. Lambda is a rate, and lambda times time must use matching units.

Worked example - Toll plaza: 50 cars/min, next car within 3 seconds

1. Random object: waiting time, so use exponential.
2. λ=50 per minute. Convert 3 seconds to 0.05 minute.
3. P(X≤0.05)=1-e^(-50×0.05).
4. =1-e^(-2.5)≈0.9179.
Result: Probability ≈0.9179.
Verify: With 50 arrivals per minute, a very short wait is quite likely, so a large probability makes sense.

KNOWLEDGE CHECK - answer before looking at the key
Calls arrive at 12 per hour. The question asks for time until the next call. Which model?

A. Exponential
B. Poisson
C. Uniform

YOUR TURN - work independently
Calls arrive at 12 per hour. Find P(next call arrives within 3 minutes). Enter 4 d.p.
Answer: _____

EXIT CRITERION
Move on when you can distinguish Poisson count from exponential wait and catch a rate/time unit mismatch before substituting.

Sources: 2026 Chapter 6 notes §§3.1-3.1.5 | Exponential probabilities worked sheet | Tutorial 6 + solutions Q10-Q13 | Chapter 6 lecturer transcript toll-plaza example

WORKSPACE / RETRIEVAL NOTES

CHAPTER 6 | LESSON 6.9 | OFFLINE GUIDED LEARN

6.9 - Inverse Exponential

OBJECTIVE
Solve for an unknown waiting-time boundary when a cumulative or tail probability is given, and keep it conceptually separate from inverse normal.

Explanation

Inverse exponential asks for time x rather than probability. Start from the exponential probability equation and solve for x using natural logarithms.

If $P(X \le x)=\alpha$, then $1-e^{(-\lambda x)}=\alpha$, so $e^{(-\lambda x)}=1-\alpha$ and $x=-\ln(1-\alpha)/\lambda$.

If the question gives a right-tail probability $q=P(X>x)$, it is even quicker: $e^{(-\lambda x)}=q$, so $x=-\ln(q)/\lambda$.

This is not NORM.INV. Exponential is asymmetric and uses ln, not z-scores.

SAY ALOUD
Inverse exponential means solve the exponential equation for time. Cumulative alpha gives x equals minus ln of one minus alpha over lambda.

Worked example - 80th percentile when λ=2 per week

1. Set $P(X<x)=0.80$.
2. $1-e^{(-2x)}=0.80$, so $e^{(-2x)}=0.20$.
3. Take ln: $-2x=\ln(0.20)$.
4. $x=-\ln(0.20)/2\approx 0.8047$ weeks.
Result: $P80\approx 0.8047$ weeks.
Verify: An 80th percentile is a positive waiting time. Substituting it back should give cumulative probability near 0.80.

KNOWLEDGE CHECK - answer before looking at the key
For cumulative probability α, which boundary formula is correct?

A. $-\ln(1-\alpha)/\lambda$
B. $\mu+z\sigma$
C. $\ln(\alpha)\lambda$

YOUR TURN - work independently
If $X\sim Exp(\lambda=3)$ and $P(X>x)=0.65$, find x.
Answer: _____

EXIT CRITERION
Move on when you can derive the inverse formula from the probability equation and know immediately that inverse normal tools do not apply.

Sources: Chapter 6 lecturer transcript inverse-exponential extension | Exponential probabilities worked sheet | 2026 Chapter 6 content architecture

WORKSPACE / RETRIEVAL NOTES

CHAPTER 6 | LESSON 6.10 | OFFLINE GUIDED LEARN

6.10 - Normal Approximation to the Binomial

OBJECTIVE
Decide when the normal approximation is allowed, construct the approximate normal model, apply the correct continuity correction, and calculate the probability.

Explanation

The binomial distribution is discrete, but for sufficiently large expected success and failure counts its shape can be approximated by a normal curve.

Current Chapter 6 conditions are $n\pi>5$ and $n(1-\pi)>5$. If either fails, do not use this approximation.

The approximating normal model has $\mu=n\pi$ and $\sigma=\sqrt{n\pi(1-\pi)}$.

Because the binomial lives on integer points while the normal curve is continuous, we use a ±0.5 continuity correction to preserve the area around each integer.

Exact x becomes $x-0.5<Xa<x+0.5$. $X\leq x$ becomes $Xa<x+0.5$. $X\geq x$ becomes $Xa>x-0.5$. Strict inequalities first need translation to the adjacent inclusive integer event.

SAY ALOUD
First check n pi and n one minus pi are both greater than five. Then mean is n pi, SD is square root n pi one minus pi, continuity-correct the integer boundary by half, then standardise.

Worked example - Tutorial 6 Q14: n=100, π=0.20, approximate P(X=25)

1. Check: $n\pi=20>5$ and $n(1-\pi)=80>5$.
2. Approximate model: $\mu=20$, $\sigma=\sqrt{16}=4$.
3. Continuity correction: $P(X=25)\approx P(24.5<Xa<25.5)$.
4. $zL=(24.5-20)/4=1.125\approx1.13$; $zU=(25.5-20)/4=1.375\approx1.38$.
5. Probability$\approx\Phi(1.38)-\Phi(1.13)=0.9162-0.8708=0.0454$.
Result: $P(X=25)\approx0.0454$.
Verify: Without the ±0.5 slice, a continuous model would assign zero probability to exactly 25.

KNOWLEDGE CHECK - answer before looking at the key
Which continuous event approximates $P(X\geq25)$?

A. $P(Xa>24.5)$
B. $P(Xa>25.5)$
C. $P(Xa<24.5)$

YOUR TURN - work independently
For $n=100$ and $\pi=0.20$, how should $P(X<25)$ be continuity-corrected?

A. $P(Xa<24.5)$
B. $P(Xa<25.5)$
C. $P(Xa>24.5)$

EXIT CRITERION
Move on when you can do the two >5 checks, construct μ and σ, and continuity-correct =, <, ≤, > and ≥ without a lookup table.

Sources: 2026 Chapter 6 notes §§1.5, pp. 14-16 | Normal approximation handwritten example | Tutorial 6 + solutions Q14-Q17 | Chapter 6 lecturer transcript overbooking / continuity-correction explanation

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7

Sampling Distributions

SCOPE
Full current Chapter 7 learning layer. The FPC bridge is retained with its source caveat.

Offline tracker: [ ] 7.1 [ ] 7.2 [ ] 7.3 [ ] 7.4 [ ] 7.5 [ ] 7.6 [ ] 7.7 [ ] 7.8 [ ] 7.9 [ ] 7.10 [ ] 7.11 [ ] 7.12 [ ] 7.13

CHAPTER 7 | LESSON 7.1 | OFFLINE GUIDED LEARN

7.1 - Population → Sample → Statistic → Sampling Distribution

OBJECTIVE
Build the core mental model: a sampling distribution is the probability distribution of a statistic over all possible random samples of a fixed size, not the raw observations in one sample.

Recognise it

The question asks what a sampling distribution is.

The question contrasts a population, one sample, one statistic and repeated possible statistics.

The wording asks why two random samples can produce different means or proportions.

Explanation

A population is the full collection of units. A sample is one subset drawn from it. A parameter such as μ, σ or π belongs to the population; a statistic such as $\bar{X}$ or p is calculated from a sample.

Different random samples usually contain different observations, so the statistic changes from sample to sample. The sampling distribution describes that variability mathematically.

The Chapter 7 objective is not merely to calculate one sample statistic; it is to reason about the distribution of the statistic before the sample is observed.

FORMULA / DECISION OBJECT
Sampling distribution = distribution of all possible values of a sample statistic for a fixed n. Random-object question: X, $\bar{X}$, p, or $\bar{X}_1$-$\bar{X}_2$?

Excel route - rehearse the syntax on paper

1. Excel can calculate a realised sample statistic, but it does not decide what the random object is for you.
2. Before typing a function, name the statistic whose repeated values form the sampling distribution.

SAY ALOUD
One sample gives me one statistic. A sampling distribution is the distribution of that statistic across all possible random samples of the same size.

Worked example - Chapter 7 GPA thought experiment

1. Imagine repeatedly selecting 50 students from the same college.
2. Each sample produces a possibly different sample mean GPA.
3. Collect the possible sample means rather than the 50 raw GPAs from one sample.
4. The distribution of those possible means is the sampling distribution of $\bar{X}$.
Result: The random object is $\bar{X}$, and the sampling distribution is the probability model for its possible values.
Verify: Do not call one realised set of 50 GPAs a sampling distribution.

COMMON TRAP
Do not confuse population distribution, one sample, one realised statistic and the sampling distribution of that statistic.

KNOWLEDGE CHECK - answer before looking at the key
Which object is a sampling distribution?

A. The distribution of all possible sample means for samples of size 25
B. The 25 observations in one sample
C. The population histogram itself

YOUR TURN - work independently
A poll repeatedly samples 100 voters and records the sample proportion supporting a candidate. What distribution is being built?

A. Sampling distribution of p
B. Population distribution of individual votes
C. Sampling distribution of $\bar{X}$

EXIT CRITERION
You can identify the population, sample, parameter, statistic and sampling distribution in a new scenario without looking.

Sources: Chapter 7 slides pp. 2-4 | Chapter 7 standalone manual: Start Here / mental model

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.2 | OFFLINE GUIDED LEARN

7.2 - Exact Sampling Distributions & Unbiasedness

OBJECTIVE
Construct an exact sampling distribution for a tiny population and use it to see why $\bar{X}$ is an unbiased estimator of μ.

Recognise it

A tiny finite population is listed explicitly.

The task says "all possible samples" or asks you to construct the sampling distribution exactly.

The question asks why the sample mean is unbiased.

Explanation

The source builds a toy population 18,20,22,24 and lists every ordered sample of size 2 with replacement. Each possible sample has its own $\bar{X}$.

An exact sampling distribution combines repeated statistic values with their probabilities. Its mean is the expected value of the statistic.

When $E(\bar{X})=\mu$, $\bar{X}$ is an unbiased estimator of μ. Unbiased does not mean every sample mean equals μ.

FORMULA / DECISION OBJECT
$\mu_{\bar{X}} = \Sigma\, \bar{X} \cdot P(\bar{X})$. Unbiasedness: $E(\bar{X})=\mu$.

Excel route - rehearse the syntax on paper

1. For a small enumeration, Excel can list sample combinations and use AVERAGE for each sample, but the key idea is the complete set of possible statistic values.
2. For Tutorial 7 Q4, sampling is without replacement, so use combinations rather than duplicating ordered draws as if replacement were allowed.

SAY ALOUD
Unbiased means the sampling distribution is centred at the true parameter. One sample mean can miss μ; the expected sample mean still equals μ.

Worked example - Source toy population 18,20,22,24

1. Population mean $\mu=(18+20+22+24)/4=21$.
2. With replacement and $n=2$ there are $4^2=16$ ordered samples.
3. Their sample means range from 18 to 24 and are no longer uniformly distributed.
4. The mean of the 16 sample means is 21.

Result: $\mu_{\bar{X}}=21=\mu$, so $\bar{X}$ is unbiased for μ.

Verify: The sampling distribution can have a different shape and smaller spread than the population while keeping the same centre.

COMMON TRAP
Do not interpret unbiasedness as "$\bar{X}$ equals μ in every sample." It is a statement about the centre across possible samples.

KNOWLEDGE CHECK - answer before looking at the key

If $E(\bar{X})=\mu$, what property has been demonstrated?

A. Unbiasedness
B. Zero sampling variation
C. Normality

YOUR TURN - work independently

Tutorial 7 Q4 population: 1,3,6,7,9,10. What is the population mean and therefore the mean of the exact sampling distribution of $\bar{X}$?

Answer: _____

EXIT CRITERION
You can enumerate a tiny-sample experiment, construct the statistic distribution and explain $E(\bar{X})=\mu$.

Sources: Chapter 7 slides pp. 5-10 | Tutorial 7 Q4 and Q10

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.3 | OFFLINE GUIDED LEARN

7.3 - Standard Error & Why Larger n Narrows the Distribution

OBJECTIVE
Treat standard error as the standard deviation of a statistic across possible samples and explain exactly how n changes it.

Recognise it

The question compares variability of sample means from different n.
The question asks for the standard error of $\bar{X}$.
The question asks why increasing sample size makes estimates more stable.

Explanation

Different samples produce different $\bar{X}$ values. The standard error measures the spread of those possible means.
For the ordinary sample-mean model (sampling with replacement, or without replacement from an effectively infinite population), $\sigma_{\bar{X}}=\sigma/\sqrt{n}$. The square root matters: quadrupling n halves the standard error. If sampling is without replacement from a finite population, the FPC in Lesson 7.12 can reduce this spread further.
Increasing n does not shift the centre $\mu_{\bar{X}}=\mu$. It narrows the distribution around the same target.

FORMULA / DECISION OBJECT
$SE(\bar{X})=\sigma/\sqrt{n}$. If n is multiplied by k, SE is divided by $\sqrt{k}$.

Excel route - rehearse the syntax on paper

1. In Excel use =sigma/SQRT(n) when population σ is known.
2. Do not use =sigma/n. SQRT(n) is the entire reason the gain from sample size has diminishing returns.

SAY ALOUD
Standard deviation describes individuals; standard error describes the statistic across samples. Larger n shrinks SE by the square-root rule, but the centre stays μ.

Worked example - Tutorial 7 Q4 comparison

1. For the population 1,3,6,7,9,10, all without-replacement n=2 sample means have mean 6.
2. Their exact sampling-distribution SD is 2.0000.
3. For all without-replacement n=3 sample means, the centre is still 6.
4. Their exact sampling-distribution SD is about 1.4142.
Result: The n=3 sampling distribution has less variability while retaining centre 6.
Verify: This exact Tutorial 7 Q4 enumeration is without replacement from N=6, so its SDs (2.0000 and 1.4142) already reflect finite-population reduction; they should not be mistaken for the ordinary $\sigma/\sqrt{n}$ values. The key comparison is still that n=3 varies less than n=2.

COMMON TRAP
Do not divide by n; divide by $\sqrt{n}$. And do not say larger n changes the population standard deviation σ. Also do not use the uncorrected $\sigma/\sqrt{n}$ formula to reproduce Tutorial 7 Q4's finite without-replacement enumeration; that case belongs to the FPC bridge.

KNOWLEDGE CHECK - answer before looking at the key
If n is multiplied by 4 while σ stays fixed, what happens to $SE(\bar{X})$?
A. It is halved
B. It is quartered
C. It is unchanged

YOUR TURN - work independently
Population σ=12 and n=36. What is $SE(\bar{X})$?
Answer: _____

EXIT CRITERION
You can distinguish SD from SE and predict the effect of any multiplicative change in n.

Sources: Chapter 7 slides pp. 9-11 and 15 | Tutorial 7 Q4(c) and Q11

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.4 | OFFLINE GUIDED LEARN

7.4 - One Observation X versus Sample Mean $\bar{X}$

OBJECTIVE
Identify whether the probability is about one observation or an average before selecting the centre and spread.

Recognise it

The wording says one person/item/time/mark versus the average of n observations.
A variance may be given and must first be converted to σ.
The task asks you to interpret a probability about $\bar{X}$ in context.

Explanation

For one observation X from a normal population, use centre μ and spread σ.
For a sample mean $\bar{X}$, use centre μ and spread σ/√n. This distinction is one of the most important Chapter 7 exam traps.
If variance is supplied, convert it to standard deviation before using either model: σ=√variance.

FORMULA / DECISION OBJECT
One X: $Z=(X-\mu)/\sigma$. Sample mean $\bar{X}$: $Z=(\bar{X}-\mu)/(\sigma/\sqrt{n})$.

Excel route - rehearse the syntax on paper

1. One normal observation: =NORM.DIST(x,mu,sigma,TRUE).
2. Sample mean: =NORM.DIST(xbar,mu,sigma/SQRT(n),TRUE).

SAY ALOUD
I name the random object first. One observation uses sigma. An average uses sigma over square-root n.

Worked example - Tutorial 7 Q2 - one mark versus class average

1. Marks are normal with μ=78 and variance 36, so σ=6.
2. Part (a) asks for one student mark below 75: z=(75-78)/6=-0.5, probability≈0.3085.
3. Part (b) asks for the mean of n=50 students below 75: SE=6/√50≈0.8485.
4. z=(75-78)/0.8485≈-3.5355, probability≈0.0002.
Result: The same boundary 75 has radically different probability because X and $\bar{X}$ have different spreads.
Verify: The population did not change; the random object changed.

COMMON TRAP
Do not use σ for a sample mean and do not use σ/√n for a single observation.

KNOWLEDGE CHECK - answer before looking at the key
Population variance is 36. A class of 50 students has an average mark below 75. Which spread belongs to the random object?

A. 6/√50
B. 36/√50
C. 6

YOUR TURN - work independently
A question asks whether the average waiting time of 36 diners exceeds 9 minutes. Which object and spread?

A. $\bar{X}$ and σ/√36
B. X and σ
C. p and √[p(1−p)/n]

EXIT CRITERION
You can classify X versus $\bar{X}$ before doing any arithmetic and interpret the result in terms of the correct object.

Sources: Chapter 7 slides pp. 11-13 | Tutorial 7 Q2 | Chapter 7 formula sheet

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.5 | OFFLINE GUIDED LEARN

7.5 - Sample-Mean Probabilities with Known σ: Left, Between & Right

OBJECTIVE
Solve forward probabilities for $\bar{X}$ when population σ is known, with the tail/region matching the words.

Recognise it

A normal population or justified normal sampling model is available.
The random object is $\bar{X}$ and population σ is known.
The task asks below, between or above a stated sample-mean boundary.

Explanation

For a normal population, the sampling distribution of $\bar{X}$ is exactly normal for every valid n, with centre μ and SE $\sigma/\sqrt{n}$.
Standardise the sample-mean boundary using the SE, not σ.
Excel cumulative functions return left area. "Above" therefore requires a complement unless you use an equivalent direct route.

FORMULA / DECISION OBJECT
$\bar{X} \sim N(\mu,\sigma^2/n)$, $Z=(\bar{X}-\mu)/(\sigma/\sqrt{n})$. Left=Φ(z); Between=Φ(zU)-Φ(zL); Right=1-Φ(z).

Excel route - rehearse the syntax on paper

1. Direct left: =NORM.DIST(xbar,mu,sigma/SQRT(n),TRUE).
2. Right: =1-NORM.DIST(xbar,mu,sigma/SQRT(n),TRUE). Between: upper cumulative minus lower cumulative.

SAY ALOUD
Known sigma and a sample mean means normal Z. I draw the region, use sigma over square-root n, and make Excel's left-area output match the requested tail.

Worked example - Tutorial 7 Q1(a-c)

1. μ=100, σ=10, n=25, so SE=10/5=2.
2. (a) $P(\bar{X}<95)$: z=-2.5, probability≈0.0062.
3. (b) $P(95<\bar{X}<97.5)$: zL=-2.5, zU=-1.25, probability≈0.0994.
4. (c) $P(\bar{X}>102.2)$: z=1.1, probability≈0.1357.
Result: Left≈0.0062; between≈0.0994; right≈0.1357.
Verify: Every part uses the same centre and SE; only the region changes.

COMMON TRAP
Do not forget the right-tail complement and do not substitute σ=10 directly into the denominator when n=25.

KNOWLEDGE CHECK - answer before looking at the key
For $\bar{X}$, "above 102.2" requires which final probability operation after standardising?
A. 1−Φ(z)
B. Φ(z)
C. Φ(zU)−Φ(zL)

YOUR TURN - work independently
For μ=100, σ=10, n=25, what is $P(\bar{X}>102.2)$?
Answer: _____

EXIT CRITERION
You can solve left, right and between $\bar{X}$ probabilities and reconstruct the Excel route without prompts.

Sources: Chapter 7 slides pp. 12-18 | Tutorial 7 Q1(a-c)

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.6 | OFFLINE GUIDED LEARN

7.6 - Inverse Sample-Mean Boundaries & Central Symmetric Limits

OBJECTIVE
Work backwards from a probability to the $\bar{X}$ boundary and construct symmetric probability limits around μ.

Recognise it

The probability is given but the sample-mean cutoff is unknown.
The wording says "there is a 65% chance $\bar{X}$ is above what value?"
The task asks for limits containing a fixed central percentage of sample means.

Explanation

Inverse problems reverse the ordinary workflow: probability → z → original $\bar{X}$ scale.
For $P(\bar{X}>x)=0.65$, the area to the left of x is 0.35, so use $z=\Phi^{-1}(0.35)$, not $\Phi^{-1}(0.65)$.
For a central C interval, split $\alpha=1-C$ equally between the two tails and transform both z boundaries back with $\mu+z\cdot SE$.

FORMULA / DECISION OBJECT
Boundary = $\mu + z\cdot\sigma/\sqrt{n}$. Central limits: $\mu \pm z^*\cdot\sigma/\sqrt{n}$, where $z^*=\Phi^{-1}((1+C)/2)$.

Excel route - rehearse the syntax on paper

1. Right-tail inverse boundary: =NORM.INV(left_area,mu,sigma/SQRT(n)).
2. Central limits: =mu ± NORM.S.INV((1+C)/2)*sigma/SQRT(n).

SAY ALOUD
Inverse means the probability comes first. I convert the requested area to left-cumulative probability, find z, then return to X-bar units.

Worked example - Tutorial 7 Q1(d)

1. μ=100, σ=10, n=25, so SE=2.
2. $P(\bar{X}>x)=0.65$ means $P(\bar{X}\leq x)=0.35$.
3. $z=\Phi^{-1}(0.35)\approx-0.3853$.
4. $x=100+(-0.3853)(2)\approx99.2294$.
Result: There is a 65% chance that $\bar{X}$ is above approximately 99.2294.
Verify: Using 0.65 as the left area would put the boundary above μ and reverse the requested region.

COMMON TRAP
Do not use the requested right-tail probability directly as a cumulative-left probability.

KNOWLEDGE CHECK - answer before looking at the key
If $P(\bar{X}>x)=0.65$, what cumulative-left area belongs to x?

A. 0.35
B. 0.65
C. 0.975

YOUR TURN - work independently
Source central-limit example: μ=368, σ=15, n=25. Enter the symmetric limits containing 95% of sample means.

EXIT CRITERION
You can convert verbal inverse/central-area wording into the correct cumulative probabilities and $\bar{X}$ boundaries.

Sources: Chapter 7 slides pp. 16-18 | Tutorial 7 Q1(d)

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.7 | OFFLINE GUIDED LEARN

7.7 - Central Limit Theorem: What Becomes Normal & When

OBJECTIVE
Use the CLT correctly for non-normal populations while preserving the sampling-distribution centre and SE.

Recognise it

The population is not stated to be normal or is explicitly skewed/exponential.
The task asks about an average of many observations.
You must decide whether the sampling distribution of $\bar{X}$ is approximately normal.

Explanation

The CLT concerns the sampling distribution of $\bar{X}$. It does not transform the original population or individual observations into normal variables.
The course guideline says normal population → exact normal $\bar{X}$ for any valid n; fairly symmetric populations may be adequate around n>15; most distributions around n>30. These are course rules of thumb, not magical theorem cutoffs.
Under the CLT, the sampling distribution remains centred at μ with SE $\sigma/\sqrt{n}$. Only the approximate shape changes toward normal.

FORMULA / DECISION OBJECT
For sufficiently large n: $\bar{X} \approx N(\mu,\sigma^2/n)$, even when the original population is not normal.

Excel route - rehearse the syntax on paper

1. Once the CLT justifies the normal sampling model, use =NORM.DIST(xbar,mu,sigma/SQRT(n),TRUE) for cumulative-left sample-mean probabilities.
2. Do not use an exponential CDF on the sample mean merely because each individual observation is exponential.

SAY ALOUD
The CLT makes the sampling distribution of the mean approximately normal. It does not make the population normal. The centre stays μ and the SE stays σ over square-root n.

Worked example - Chapter 7 restaurant waiting-time example

1. Individual wait times have μ=8 min and σ=3 min; their population shape need not be normal.
2. n=36, so the course CLT guideline supports an approximately normal $\bar{X}$ distribution.
3. $SE=3/\sqrt{36}=0.5$.
4. $P(7.8<\bar{X}<8.2)$: $z_L=-0.4$, $z_U=0.4$.
5. $\Phi(0.4)-\Phi(-0.4)=0.6554-0.3446=0.3108$.
Result: Approximately 0.3108 of samples of 36 have mean wait between 7.8 and 8.2 minutes.
Verify: The statement is about sample means, not individual waiting times.

COMMON TRAP
Do not say "n=30 makes the population normal." Also do not forget to convert the exponential rate to consistent time units before finding μ and σ.

KNOWLEDGE CHECK - answer before looking at the key
What becomes approximately normal under the CLT?
A. The sampling distribution of $\bar{X}$
B. The original population itself
C. Every raw observation

YOUR TURN - work independently
Tutorial 7 Q3(b): interarrival times are exponential with rate 400/hour (mean 9 sec). For n=30, approximate $P(\bar{X}>12$ sec).
Answer: _____

EXIT CRITERION
You can state exactly what the CLT changes, apply the course size guidance and execute a non-normal sample-mean probability.

Sources: Chapter 7 slides pp. 19-24 | Tutorial 7 Q3(b) | Chapter 7 standalone manual CLT section

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.8 | OFFLINE GUIDED LEARN

7.8 - Sample Mean with σ Unknown: Student t

OBJECTIVE
Use the current Chapter 7 method fork: known population σ → Z; unknown σ with sample s → Student t, df=n-1.

Recognise it

The random object is $\bar{X}$.

Population σ is explicitly unknown while sample standard deviation s is supplied or computed.

The task asks for a probability and the current source directs you to Student t / T.DIST.

Explanation

The signed-off current Chapter 7 source map verifies a Student-t route in §2.6: replace unknown σ with sample s, standardise with $T=(\bar{X}-\mu)/(s/\sqrt{n})$, and use df=n-1.

The t distribution accounts for the extra uncertainty created by estimating σ with s.

This route is different from the known-σ Z model even if n is supplied. The method fork is driven by σ status, not by a reflexive "large n means Z" shortcut.

FORMULA / DECISION OBJECT
Unknown σ: $T=(\bar{X}-\mu)/(s/\sqrt{n})$, df=n-1. Current Excel cumulative-left route: =T.DIST(tcalc,df,TRUE).

Excel route - rehearse the syntax on paper

1. Compute tcalc from the sample mean and sample s.
2. Use =T.DIST(tcalc,df,TRUE) for cumulative-left probability under the current §2.6 route; use appropriate complements/differences for other regions.

SAY ALOUD
Sample mean plus unknown population sigma means Student t. I use s, divide by square-root n, use df n minus one, then make the t cumulative area match the requested region.

Worked example - 2026 A1S2 practice-test trap

1. The practice test describes a normally distributed population but says population σ is unknown.
2. n=25 and sample s=4, so df=24 and SE=4/√25=0.8.
3. For $18.5<\bar{X}<19.5$ around μ=19, t limits are -0.625 and +0.625.
4. The Student-t probability is approximately 0.4621.
5. The Z route gives about 0.4680, which appears as a distractor.
Result: Use Student t; the correct practice-test option is approximately 0.4621, not the Z distractor 0.4680.
Verify: This is direct evidence that distribution selection matters in the A1S2 style.

COMMON TRAP
Do not choose Z just because the sample is "large enough." First ask whether population σ is known.

KNOWLEDGE CHECK - answer before looking at the key
Population σ is unknown, sample s is supplied, n=25. Which route is source-aligned?

A. Student t with df=24
B. Z because n is bigger than 20
C. Normal with denominator σ

YOUR TURN - work independently
$\bar{X}=54$, μ=50, s=10, n=25. Calculate T.
Answer: _____

EXIT CRITERION
You can run the known-σ versus unknown-σ fork automatically, compute t and df, and reconstruct the Excel cumulative route.

Sources: Current signed-off Chapter 7 source map: SDS Chapter 7 NOTES.pdf §2.6 | 2026 A1S2 Practice Test Q8 | A1S2 2026 formula sheet inference rows

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.9 | OFFLINE GUIDED LEARN

7.9 - Sample Proportion: p, π, Conditions & Standard Error

OBJECTIVE
Build the sampling model for p, distinguish sample p from population π, check the approximation conditions, and choose theoretical versus estimated SE correctly.

Recognise it

The data classify observations as success/failure or yes/no.

The question asks about a sample percentage/proportion.

Known π is supplied for a sampling probability, or observed p is used when an estimated SE is specifically requested.

Explanation

p=X/n is the sample proportion; π is the fixed population proportion. The sampling distribution of p is centred at π.

The Chapter 7 deck uses the approximate-normal conditions nπ≥5 and n(1-π)≥5; the current master applies the stricter current wording >5 where explicitly patched. Preserve the current system rule when practising in this master.

For a theoretical sampling probability with known π, SE=√[π(1-π)/n]. When an estimated SE is requested and π is unknown, use √[p(1-p)/n].

FORMULA / DECISION OBJECT
p=X/n; μp=π; theoretical SE(p)=√[π(1-π)/n]. Estimated SE=√[p(1-p)/n] when estimation context requests it.

Excel route - rehearse the syntax on paper

1. Sample p = successes/n.
2. Theoretical SE: =SQRT(pi*(1-pi)/n). Estimated SE: =SQRT(p*(1-p)/n) only in the appropriate estimation context.

SAY ALOUD
p is my sample statistic and pi is my population parameter. Known pi drives the theoretical sampling SE; observed p is used only when I am estimating the SE.

Worked example - Tutorial 7 Q5

1. n=64 and successes X=48, so p=48/64=0.75.
2. The population proportion is known: π=0.70.
3. Theoretical SE=√[0.70(0.30)/64]≈0.05728.
4. The sample p need not equal π; it varies from sample to sample.
Result: p=0.75 and theoretical SE≈0.0573.
Verify: Do not substitute p=.75 into the theoretical SE when the probability model gives π=.70.

COMMON TRAP
Do not confuse π with p, and do not switch between theoretical and estimated SE without reading what the task asks.

KNOWLEDGE CHECK - answer before looking at the key
n=64, 48 successes, known population π=0.70. What is the sample proportion?

A. 0.75
B. 0.70
C. 0.0573

YOUR TURN - work independently
Tutorial 7 Q6: 14 of 40 students answer yes. Enter p and the estimated SE requested by the question.

EXIT CRITERION
You can compute p, identify π, check the approximation rule and choose the correct SE form.

Sources: Chapter 7 slides pp. 25-28 | Tutorial 7 Q5-Q6 | Current master proportion-condition patch

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.10 | OFFLINE GUIDED LEARN

7.10 - Proportion Probabilities, Between Regions & Inverse Bounds

OBJECTIVE
Use the normal sampling model for p to solve forward probabilities, between regions and central inverse bounds, and explain how n changes precision.

Recognise it

Known π and n are supplied and the question asks about a sample percentage.
The event is above/below/between p boundaries.
The probability is given and the task asks for symmetric limits around π.

Explanation

Once the approximation conditions pass, standardise p using z=(p-π)/SE with SE based on known π.
Between probabilities require two standardised boundaries and subtraction of cumulative areas.
For central C bounds, use $z^*=\Phi^{-1}((1+C)/2)$ and π±z*SE. Increasing n reduces SE and therefore narrows the distribution/bounds.

FORMULA / DECISION OBJECT
$Z=(p-\pi)/\sqrt{[\pi(1-\pi)/n]}$. Central p bounds: $\pi \pm z^*\sqrt{[\pi(1-\pi)/n]}$.

Excel route - rehearse the syntax on paper

1. Left p probability: =NORM.DIST(p_boundary,pi,SQRT(pi*(1-pi)/n),TRUE).
2. Between = upper cumulative - lower cumulative; central bounds = pi ± NORM.S.INV((1+C)/2)*SE.

SAY ALOUD
For a sample proportion probability, I centre at pi, use the theoretical proportion SE, sketch the p region, and then standardise.

Worked example - Chapter 7 deck example

1. π=0.40, n=200.
2. SE=√[.4(.6)/200]≈0.03464.
3. P(.40≤p≤.45): zL=0, zU≈1.44.
4. Using the source's rounded z=1.44 and cumulative table gives 0.9251-0.5000=0.4251. Using full precision in Excel (z≈1.44338) gives ≈0.4255; the difference is rounding route, not a different model.
Result: Source table route: P(.40≤p≤.45)≈0.4251 after rounding z to 1.44. Full-precision Excel route: ≈0.4255.
Verify: The object is p, not the underlying binomial count X. In Excel keep full precision rather than forcing the table-rounded 0.4251.

COMMON TRAP
Do not apply continuity correction to the p model used in this Chapter 7 sampling-distribution route; that belongs to the Chapter 6 binomial-count approximation problem.

KNOWLEDGE CHECK - answer before looking at the key
If n increases while π stays fixed, what happens to the sampling distribution of p?
A. It narrows around the same centre π
B. Its centre moves toward 0
C. Its SE increases

YOUR TURN - work independently
Tutorial 7 Q8(a): π=0.20, n=100. Approximate P(0.15<p<0.25).
Answer: _____

EXIT CRITERION
You can solve forward and inverse p questions and explain the effect of larger n on the distribution.

Sources: Chapter 7 slides pp. 26-31 | Tutorial 7 Q7-Q9 | Chapter 7 formula sheet

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.11 | OFFLINE GUIDED LEARN

7.11 - Difference Between Two Independent Sample Means

OBJECTIVE
Treat $\bar{X}_1-\bar{X}_2$ as one random variable, preserve group order, combine variances correctly and translate comparison wording into a threshold.

Recognise it

Two independent populations/groups are compared using sample averages.
Population means, population SDs and sample sizes are supplied.
The wording says group 1 exceeds group 2, or exceeds it by more than c.

Explanation

For the sourced known-σ independent case, the difference is centred at $\mu_1-\mu_2$. It is not automatically centred at zero.
Independent mean variances add: $SE=\sqrt{(\sigma_1^2/n_1+\sigma_2^2/n_2)}$. Do not add the standard deviations directly.
Fix the order first. "Group 1 exceeds group 2" means $\bar{X}_1-\bar{X}_2>0$; "by more than 25" means $\bar{X}_1-\bar{X}_2>25$.

FORMULA / DECISION OBJECT
$\mu diff=\mu_1-\mu_2$; $SEdiff=\sqrt{(\sigma_1^2/n_1+\sigma_2^2/n_2)}$; $Z=[(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)]/SEdiff$.

Excel route - rehearse the syntax on paper

1. Compute SE with =SQRT(sigma1^2/n1 + sigma2^2/n2).
2. For a boundary c, use =1-NORM.DIST(c,mu1-mu2,SE,TRUE) for $P(\bar{X}_1-\bar{X}_2>c)$.

SAY ALOUD
I freeze the order one minus two, centre the difference at mu one minus mu two, add variances inside the square root, then translate the wording to a threshold.

Worked example - Source MBA starting-salary example

1. US: $\mu_1=620000$, $\sigma_1=145000$, $n_1=50$; UCT: $\mu_2=600000$, $\sigma_2=183000$, $n_2=60$.
2. Centre $\mu diff=20000$.
3. $SE=\sqrt{(145000^2/50+183000^2/60)}\approx31283.3822$.
4. Event US sample mean exceeds UCT: $\bar{X}_1-\bar{X}_2>0$.
5. $z=(0-20000)/31283.3822\approx-0.6393$; right-tail$\approx0.7387$.
Result: $P(\bar{X}US>\bar{X}UCT)\approx0.7387$.
Verify: The centre is +20000 because the order is US-UCT.

COMMON TRAP
Do not reverse the groups because an observed mean is larger, and do not centre the difference at zero unless $\mu_1-\mu_2$ actually equals zero.

KNOWLEDGE CHECK - answer before looking at the key
"Mean 1 exceeds mean 2 by more than 25" translates to which event?
A. $\bar{X}_1-\bar{X}_2>25$
B. $\bar{X}_2-\bar{X}_1>25$
C. $\bar{X}_1+\bar{X}_2>25$

YOUR TURN - work independently
Tutorial 7 Q12(a): μ1=280,σ1=25, μ2=270,σ2=30, n1=n2=10. Find $P(\bar{X}1-\bar{X}2>25)$.
Answer: _____

EXIT CRITERION
You can preserve order, build the difference model, translate thresholds and calculate a probability independently.

Sources: Chapter 7 slides pp. 32-38 | Tutorial 7 Q12-Q13 | A1S2 formula sheet two-independent-means row

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.12 | OFFLINE GUIDED LEARN

7.12 - Finite Population Correction: Without-Replacement Bridge

OBJECTIVE
Recognise when sampling without replacement from finite N changes the standard error, while preserving the current system's provisional scope flag for these Chapter 7 skills.

Recognise it

A finite population size N is supplied.

Sampling is explicitly without replacement.

The sampling fraction n/N exceeds 5% under the course rule.

Explanation

The Chapter 7 deck includes an online finite-population section: when sampling without replacement and $n/N>0.05$, multiply the usual SE by $FPC=\sqrt{[(N-n)/(N-1)]}$.

FPC reduces the spread because sampling a substantial part of a finite population leaves less uncertainty. It does not move the centre.

The canonical master still marks C7.52-C7.58 PROVISIONAL as a scope safeguard. This Guided Learn lesson teaches the sourced bridge without silently promoting those mastery skills.

FORMULA / DECISION OBJECT
$FPC=\sqrt{[(N-n)/(N-1)]}$. Mean $SE=\sigma/\sqrt{n} \times FPC$. Proportion $SE=\sqrt{[p(1-p)/n]} \times FPC$.

Excel route - rehearse the syntax on paper

1. =SQRT((N-n)/(N-1)) computes the correction factor.
2. Multiply the ordinary mean/proportion SE by that factor before standardising.

SAY ALOUD
Finite population correction needs finite N, sampling without replacement and a meaningful sampling fraction. It narrows the SE; it does not change the centre.

Worked example - Tutorial 7 Q14

1. Population of Ping-Pong balls: $\mu=1.30$, $\sigma=0.04$, $N=200$.
2. $n=16$ without replacement, so $n/N=0.08>0.05$ and FPC applies.
3. $FPC=\sqrt{(184/199)}\approx0.96157$; $SE=.04/4\times FPC\approx0.009616$.
4. $P(1.31<\bar{X}<1.33)$: $z_L\approx1.03996$, $z_U\approx3.11989$.
5. Probability≈0.1483.
Result: $P(1.31<\bar{X}<1.33)\approx0.1483$ under the sourced FPC route.
Verify: Without FPC the spread would be slightly larger; the centre remains 1.30 either way.

COMMON TRAP
Do not use FPC with replacement, and do not let this teaching lesson silently convert the existing PROVISIONAL mastery status into ACTIVE evidence.

KNOWLEDGE CHECK - answer before looking at the key

$N=200$, $n=16$, sampling without replacement. Under the source rule, apply FPC?

A. Yes, because $n/N=0.08>0.05$
B. No, because $n<30$
C. No, FPC is only for proportions

YOUR TURN - work independently

For $N=1000$ and $n=100$ without replacement, enter the FPC $\sqrt{[(N-n)/(N-1)]}$.

Answer: _____

EXIT CRITERION
You can identify the FPC trigger, calculate the corrected SE and explain what the correction changes.

Sources: Chapter 7 slides online finite-population section pp. 40-41 | Tutorial 7 Q14 | Chapter 8 finite-population source as downstream reinforcement

WORKSPACE / RETRIEVAL NOTES

CHAPTER 7 | LESSON 7.13 | OFFLINE GUIDED LEARN

7.13 - Chapter 7 Synthesis: Object → Model → SE → Region → Excel

OBJECTIVE
Select the correct Chapter 7 route before calculation and connect the chapter to the Excel-first A1S2 workflow.

Recognise it

The question is deliberately unlabelled.

You must decide whether the object is X, $\bar{X}$, p or $\bar{X}_1$-$\bar{X}_2$.

You must decide known σ versus unknown σ, CLT/normality, proportion conditions or independent-two-mean structure.

Explanation

Every Chapter 7 calculation becomes easier after you identify the random object. The object determines the centre and standard error.

For a sample mean, ask whether population σ is known. Known σ → normal/Z route. Unknown σ → Student t route in the current source. For p, check the approximation conditions. For two means, preserve group order and independence.

Then draw/translate the requested region, execute the Excel function if the A1S2 workflow calls for it, and interpret the result in terms of the statistic-not individual observations.

FORMULA / DECISION OBJECT
Decision spine: Object → centre → SE → conditions/model → region → standardise → cumulative/inverse calculation → contextual interpretation.

Excel route - rehearse the syntax on paper

1. $\bar{X}$ known σ: NORM.DIST / NORM.INV with σ/SQRT(n). $\bar{X}$ unknown σ: T.DIST with tcalc and df.
2. p: NORM.DIST with centre π and theoretical SE. Difference: NORM.DIST on the difference distribution.

SAY ALOUD
I do not start with a formula. I start by naming what is random, then I write its centre and standard error, check conditions, draw the region, and only then calculate.

Worked example - Four-way method fork

1. One observation X → centre μ, spread σ.
2. Sample mean $\bar{X}$ with known σ → centre μ, SE σ/√n, Z/normal route.
3. Sample mean $\bar{X}$ with unknown σ and sample s → Student t, df=n-1.
4. Sample proportion p → centre π, proportion SE and approximation conditions.
5. Difference $\bar{X}_1$-$\bar{X}_2$ → centre μ_1-μ_2, combined-variance SE, preserved group order.
Result: The random object determines the model; Excel executes the selected route rather than selecting it for you.
Verify: If you cannot name the random object before calculating, return to Lesson 7.1/7.4.

COMMON TRAP
Do not let a familiar Excel function conceal a wrong random object or wrong distribution.

KNOWLEDGE CHECK - answer before looking at the key
A question asks for the probability that a sample average exceeds a boundary, population σ is unknown and s is supplied. What is the route?

A. $\bar{X}$ → Student t
B. X → normal with σ
C. p → normal approximation

YOUR TURN - work independently
Which sequence is safest in an A1S2-style problem?

A. Identify object → choose model/SE → translate region → execute in Excel → interpret
B. Open Excel → try functions until one returns a number
C. Use Z whenever n is large

EXIT CRITERION
You can route an unlabelled Chapter 7 problem correctly and execute it without lesson support.

Sources: Chapter 7 slides full chapter summary | Tutorial 7 Q1-Q14 | Current Student-t §2.6 source map | 2026 A1S2 practice-test execution evidence

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8

Confidence Interval Estimation

SCOPE
C8.01-C8.68 core is included. C8.69-C8.75 auditing/population-total extension remains HOLD and is excluded from ordinary A1S2 study.

Offline tracker: [ ] 8.1 [ ] 8.2 [ ] 8.3 [ ] 8.4 [ ] 8.5 [ ] 8.6 [ ] 8.7 [ ] 8.8 [ ] 8.9 [ ] 8.10 [ ] 8.11 [ ] 8.12 [ ] 8.13

CHAPTER 8 | LESSON 8.1 | OFFLINE GUIDED LEARN

8.1 - Point Estimates, Interval Estimates & Good Estimators

OBJECTIVE
Distinguish a population parameter from its sample estimator, distinguish point from interval estimation, and explain unbiasedness, consistency and relative efficiency in the terminology used by the Chapter 8 slides.

Recognise it

The question asks what one sample statistic estimates.

The question contrasts a single estimate with a range.

The question asks which estimator is unbiased, consistent or relatively more efficient.

Explanation

A population parameter is an unknown population characteristic such as the mean μ or proportion π. A sample statistic such as $\bar{X}$ or p is used to estimate it.

A point estimate is one number. A confidence interval adds information about precision by giving lower and upper confidence limits around a point estimate.

An unbiased estimator has expected value equal to the population parameter. A consistent estimator gets closer to the parameter as sample size increases. Among two unbiased estimators, the one with smaller variance is relatively more efficient.

FORMULA / DECISION OBJECT
Population mean μ ← estimated by $\bar{X}$; population proportion π ← estimated by p. Point estimate = one statistic; interval estimate = point estimate plus quantified uncertainty.

Excel route - rehearse the syntax on paper

1. No special Excel function is needed for these definitions. In workbook questions, identify the statistic you are calculating before building any formula.
2. If raw values are supplied and the point estimate is a mean, AVERAGE(range) produces $\bar{X}$; if the point estimate is a proportion, compute successes / n.

SAY ALOUD
The parameter belongs to the population; the estimator comes from the sample. A point estimate is one number. A confidence interval adds uncertainty. Unbiased means centred correctly, consistent means improving with larger n, and relative efficiency means smaller variance among unbiased estimators.

Worked example - Source estimation-process diagram

1. The population mean μ is unknown.
2. A random sample is drawn from the population.
3. The sample mean $\bar{X}$=50 is the point estimate of μ.
4. An interval such as 40 to 60 adds a quantified range around the estimate.

Result: $\bar{X}$ is the point estimator; the confidence interval is the interval estimate for μ.

Verify: Do not describe a confidence interval as an interval for individual observations.

COMMON TRAP
Do not swap the population parameter with the sample statistic, and do not treat a confidence interval for a parameter as a range for individual observations.

KNOWLEDGE CHECK - answer before looking at the key
Two estimators are both unbiased for the same parameter. Estimator A has variance 4 and estimator B has variance 9. Which is relatively more efficient?

A. Estimator A
B. Estimator B
C. They are equally efficient

YOUR TURN - work independently
Which statement matches consistency?

A. The estimator tends to get closer to the parameter as n increases.
B. The estimator is exactly equal to the parameter in every sample.
C. The estimator must have variance zero.

EXIT CRITERION
You can name the parameter, its estimator, point versus interval estimation, and all three estimator characteristics without looking.

Sources: Chapter 8 slides pp. 4-9, especially pp. 5-7 and p. 13 | Chapter 8 slide objectives/outline pp. 2-3

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.2 | OFFLINE GUIDED LEARN

8.2 - Confidence Level, α, Margin of Error & Correct Interpretation

OBJECTIVE
Understand what a confidence interval says, connect confidence level to α and α/2, identify margin of error, and avoid the fixed-interval probability mistake.

Recognise it

The question asks what "95% confidence" means.

The question asks for α or α/2 from a confidence level.

The question asks what widens or narrows an interval, or what the margin of error represents.

Explanation

The Chapter 8 source gives the general structure: point estimate ± (critical value)(standard error). The amount added and subtracted is the margin of error.

For a 95% confidence level, 1-α=0.95, so α=0.05 and α/2=0.025 in each tail for a two-sided interval.

The long-run interpretation is procedural: 95% of intervals constructed in this way would contain the unknown parameter. Once one particular interval has been calculated, the fixed parameter either lies in it or it does not.

FORMULA / DECISION OBJECT
Confidence interval = Point estimate ± Margin of error, where Margin of error = Critical value × Standard error.

Excel route - rehearse the syntax on paper

1. Excel may calculate the critical value and standard error, but you must still identify the confidence coefficient, α and whether the interval is two-sided.
2. For a 95% two-sided normal interval, the source uses $z_{\alpha/2}=1.96$.

SAY ALOUD
Confidence describes the long-run interval procedure, not a probability that the fixed parameter moves around. For 95%, alpha is 0.05 and each tail gets 0.025.

Worked example - Cereal-fill confidence interpretation

1. The source considers repeated samples of the same size.
2. Each sample gives a different $\bar{X}$ and therefore a different interval.
3. Most intervals produced by the 95% procedure contain μ; some do not.
4. In practice only one sample is observed, so we report that we are 95% confident that its interval contains μ.
Result: Confidence is a property of the repeated procedure used to construct the interval.
Verify: Never reinterpret a confidence interval as the range containing 95% of individual data values.

COMMON TRAP
Do not say there is a 95% probability that the fixed parameter lies in the one interval already computed, and do not say 95% of individuals lie inside the CI.

KNOWLEDGE CHECK - answer before looking at the key
After a 95% confidence interval has been calculated, which statement is source-aligned?

A. We are 95% confident using a procedure that captures the parameter in 95% of such intervals.
B. There is a 95% probability that the fixed parameter now lies in this one interval.
C. 95% of individual observations must lie inside the interval.

YOUR TURN - work independently
A two-sided confidence interval has confidence level 98%. What is α/2? Enter a decimal.
Answer: _____

EXIT CRITERION
You can move from confidence level → α → α/2 and can give the correct long-run interpretation aloud.

Sources: Chapter 8 slides pp. 8-21, especially pp. 14-20 | Tutorial 8 Q3 (confidence-interval interpretation)

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.3 | OFFLINE GUIDED LEARN

8.3 - Confidence Interval for μ when Population σ is Known

OBJECTIVE
Recognise the known-σ mean interval, check the source assumptions, calculate zα/2, standard error, margin of error and limits, then interpret the interval in context.

Recognise it

The target parameter is the population mean μ.

The population standard deviation σ is explicitly known.

The population is normal, or the source rule permits a large sample when it is not normal.

Explanation

When σ is known, the source uses the standard normal critical value zα/2 and standard error σ/√n.

The source states the assumptions as: σ known; population normally distributed; if the population is not normal, use a large sample n>30.

Higher confidence uses a larger critical value and therefore a wider interval. Larger n reduces standard error and narrows the interval.

FORMULA / DECISION OBJECT
$\bar{X} \pm z\alpha/2(\sigma/\sqrt{n})$

Excel route - rehearse the syntax on paper

1. Build the standard error as =sigma/SQRT(n).
2. For a two-sided confidence level stored as CL, obtain a positive critical value with =NORM.S.INV(1-(1-CL)/2) or use the equivalent source-style absolute value of NORM.S.INV((1-CL)/2).
3. Lower = Xbar - z*SE; Upper = Xbar + z*SE.

SAY ALOUD
Mean parameter, sigma known: use Z. Standard error is sigma over root n. Interval is X-bar plus or minus z alpha over two times that standard error.

Worked example - Tutorial 8 Q1: $\bar{X}$=85, σ=8, n=64, 95% confidence

1. Target: μ. Population σ is known, so select the Z interval.
2. 95% confidence gives zα/2=1.96.
3. SE=σ/√n=8/√64=1.
4. Margin=1.96×1=1.96.
5. Limits: 85-1.96=83.04 and 85+1.96=86.96.
Result: 95% CI: (83.04, 86.96).
Verify: The interval is centred at $\bar{X}$=85 and is symmetric because the same margin is added and subtracted.

COMMON TRAP
Do not use Student t merely because a sample is involved: when population sigma is explicitly known, the source route is Z. Also do not forget the square root of n in the standard error.

KNOWLEDGE CHECK - answer before looking at the key
A question asks for a confidence interval for μ and explicitly gives population σ. Which critical distribution is used?

A. Standard normal Z
B. Student t
C. Exponential

YOUR TURN - work independently
Tutorial 8 Q2: $\bar{X}$=125, σ=24, n=36. Construct the 99% confidence interval for μ. Enter lower and upper limits.

EXIT CRITERION
You can recognise known σ instantly and calculate the complete interval from confidence level to interpretation without prompts.

Sources: Chapter 8 slides pp. 17-24 | Tutorial 8 Q1, Q2 and Q4 | A1S2 2026 formula sheet: population mean, σ known inference row

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.4 | OFFLINE GUIDED LEARN

8.4 - Student t Distribution, Degrees of Freedom & the Supplied t Table

OBJECTIVE
Explain why Student t replaces Z when σ is unknown, calculate df=n-1, understand the shape of the t family, and read the exact supplied t table correctly.

Recognise it

Population σ is unknown and sample s is used instead.

A critical t value is requested from confidence level and n.

The question gives a t table and asks you to locate a row/column.

Explanation

Replacing unknown σ with sample s introduces extra uncertainty because s varies from sample to sample. The source therefore uses Student t.

Student t is a family of symmetric bell-shaped distributions with fatter tails than the standard normal. Its critical value depends on degrees of freedom, df=n-1.

The supplied table contains t values in the body. Choose the df row, then use the one-tail/two-tail probability heading that matches the confidence interval. As df grows, t approaches Z.

FORMULA / DECISION OBJECT
df = n-1; for a two-sided (1-α) confidence interval use the t critical value with upper-tail area α/2.

Excel route - rehearse the syntax on paper

1. The Chapter 8 Excel example uses =T.INV.2T(1-confidence_level, df) to obtain the positive two-sided critical t value.
2. For a 95% interval and n=100, df=99 and the source uses t≈1.9842.

SAY ALOUD
Sigma unknown means I pay for the extra uncertainty with Student t. Degrees of freedom are n minus one. For a two-sided confidence interval I use alpha over two in each tail.

Worked example - Tutorial 8 Q9(a): 95% confidence, n=10

1. Compute df=n-1=9.
2. 95% confidence gives α=.05 and α/2=.025.
3. Go to df=9 in the supplied t table.
4. Read the one-tail .025 / cumulative .975 column.
5. The table gives t=2.262.
Result: t critical = 2.262.
Verify: The table body contains t values, not probabilities; the probability is selected from the heading.

COMMON TRAP
Do not use df=n, and do not read a probability out of the body of the supplied t table. The body contains t values; choose the correct df row and tail-probability column.

KNOWLEDGE CHECK - answer before looking at the key
For n=10 and a 95% two-sided confidence interval, which row and tail area should you use in the supplied t table?

A. df=9 and one-tail 0.025
B. df=10 and one-tail 0.05
C. df=9 and one-tail 0.05

YOUR TURN - work independently
Tutorial 8 Q9(e): 90% confidence and n=16. Find the positive t critical value from the supplied table.
Answer: _____

EXIT CRITERION
You can read the exact supplied t table without guessing a column and can explain why t approaches Z as df increases.

Sources: Chapter 8 slides pp. 25-36, especially pp. 27-33 | Supplied t-table.pdf / SDS 188 CHAPTER 8 t-table.pdf | Tutorial 8 Q9

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.5 | OFFLINE GUIDED LEARN

8.5 - Confidence Interval for μ when Population σ is Unknown

OBJECTIVE
Construct and interpret a mean confidence interval when σ is unknown using s, Student t and df=n-1, including the source-supported normality check and Excel route.

Recognise it

The target is μ.

Population σ is unknown; sample standard deviation s is supplied.

The source asks you to assume/check an approximately normal population for the t interval.

Explanation

Use the t interval $\bar{X} \pm t\alpha/2(s/\sqrt{n})$. The centre is still $\bar{X}$, but the critical value comes from Student t with df=n-1.

The source says interpreting the interval requires an approximately normal population and suggests checking this with a normal probability plot or boxplot.

A confidence interval for μ estimates the population mean, not the range of individual observations. That is why an individual value may lie outside the CI without being unusual.

FORMULA / DECISION OBJECT
$\bar{X} \pm t\alpha/2,df\ (s/\sqrt{n})$, with df=n-1.

Excel route - rehearse the syntax on paper

1. SE: =s/SQRT(n).
2. df: =n-1.
3. Critical t: =T.INV.2T(1-confidence_level,df).
4. Half width = t*SE; lower = Xbar-halfwidth; upper = Xbar+halfwidth.

SAY ALOUD
Mean parameter, sigma unknown: use s and Student t with n minus one degrees of freedom. The interval estimates the population mean, not individual observations.

Worked example - Source travel-time example: n=100, $\bar{X}$=110.27, s=28.95, 95% confidence

1. σ is unknown, so use Student t.
2. df=99; the source gives tα/2=1.9842.
3. SE=28.95/√100=2.895.
4. Margin=1.9842×2.895≈5.7443.
5. Limits are approximately 104.53 and 116.01.
Result: 95% CI: 104.53 ≤ μ ≤ 116.01.
Verify: The interval is for μ. The source separately reminds you to assess approximate normality.

COMMON TRAP
Do not switch back to Z just because n looks large when population sigma is unknown in this Chapter 8 route; use sample s and Student t with df=n-1. The interval estimates mu, not an individual value.

KNOWLEDGE CHECK - answer before looking at the key
Why does the source switch from Z to t when σ is unknown?

A. Because replacing σ with variable sample s adds uncertainty.
B. Because t is used for every sample size regardless of σ.
C. Because $\bar{X}$ is no longer a mean.

YOUR TURN - work independently
Tutorial 8 Q10(a): n=24, $\bar{X}$=141.3, s=19.1. Assuming normality, construct a 90% confidence interval for μ.

EXIT CRITERION
You can choose t from the wording, determine df, calculate the interval, and explain why the CI is about μ rather than one tyre or one observation.

Sources: Chapter 8 slides pp. 27-36 | Tutorial 8 Q9-Q10 | Supplied t table | A1S2 2026 formula sheet: population mean, σ unknown inference row

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.6 | OFFLINE GUIDED LEARN

8.6 - Confidence Interval for the Population Proportion π

OBJECTIVE
Use the sample proportion p to estimate π, check the source normal-approximation condition, calculate the estimated standard error and construct/interpret a z confidence interval.

Recognise it

The variable is binary/categorical and the target is a population proportion π.
You are given successes x and sample size n, or sample p and n.
The task is a confidence interval rather than a hypothesis test.

Explanation

Compute p=x/n. For interval estimation, the source estimates the standard error with √[p(1-p)/n].
The Chapter 8 slide states the large-sample condition as np>5 and n(1-p)>5.
The confidence interval is p ± zα/2√[p(1-p)/n]. Interpret the limits as plausible values for the population proportion π under the confidence procedure.

FORMULA / DECISION OBJECT
p ± zα/2 √[p(1-p)/n]

Excel route - rehearse the syntax on paper

1. Sample proportion: =successes/n.
2. Critical z: =NORM.S.INV(1-(1-confidence_level)/2).
3. SE: =SQRT(p*(1-p)/n).
4. Half width = z*SE; lower/upper = p ± half width.

SAY ALOUD
For a proportion interval, p estimates pi. I use p inside the estimated standard error, check np and n one-minus-p, then use the Z critical value.

Worked example - Source left-handed example: 25 of 100, 95% confidence

1. p=25/100=.25.
2. Check np=25>5 and n(1-p)=75>5.
3. SE=√(.25×.75/100)=.0433.
4. Margin=1.96×.0433≈.0849.
5. Limits: .25-.0849=.1651 and .25+.0849=.3349.
Result: 95% CI for π: (0.1651, 0.3349).
Verify: The interval is for the population proportion, not for 95% of people.

COMMON TRAP
Do not use a null-hypothesis proportion inside a confidence-interval standard error. For the Chapter 8 proportion CI, use the observed sample p in the estimated SE.

KNOWLEDGE CHECK - answer before looking at the key
For a confidence interval for π, which quantity belongs inside the estimated standard error?
A. Sample proportion p
B. Null proportion π_0
C. Population mean μ

YOUR TURN - work independently
Tutorial 8 Q8(a): 303 of 326 business people reported an interruption. Construct the 95% confidence interval for π.

EXIT CRITERION
You can compute p, verify the condition, build the interval and state the interpretation in terms of π.

Sources: Chapter 8 slides pp. 37-44 | Tutorial 8 Q8 | A1S2 Practice Test Q2 (proportion CI execution style)

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.7 | OFFLINE GUIDED LEARN

8.7 - Required Sample Size for a Population Mean

OBJECTIVE
Determine the required sample size for estimating μ at a chosen confidence and margin of error, understand the role of σ, and always round the final requirement upward.

Recognise it

The question asks "how large a sample" or "minimum required n".

The target is a population mean.

A margin of error e, confidence level and σ (or an estimate of σ) are supplied.

Explanation

The source defines margin of error e as the amount added and subtracted to the point estimate. Starting from $e=z\alpha/2\sigma/\sqrt{n}$ and solving for n gives $n=z^2\sigma^2/e^2$.

To determine n you need the confidence level, acceptable margin of error and σ. If σ is unknown, the source says use a conservative anticipated value or estimate σ from a pilot sample.

Required sample size is always rounded upward because rounding downward would fail to guarantee the requested precision.

FORMULA / DECISION OBJECT
$n = z\alpha/2^2\ \sigma^2 / e^2$; always round up.

Excel route - rehearse the syntax on paper

1. Critical z from confidence level with NORM.S.INV.
2. Calculated n can be built as =(z*sigma/e)^2.
3. Required n: =ROUNDUP(calculated_n,0).

SAY ALOUD
Sample size for a mean comes from the margin-of-error equation. Smaller error or higher confidence needs a larger n, and I always round the required n up.

Worked example - Source download-time example: σ=25, e=5, 95% confidence

1. 95% gives z=1.96.
2. $n=(1.96\times25/5)^2$.
3. n=96.04.
4. Round upward.
Result: Required n=97.
Verify: If you report 96, the requested maximum margin of error is no longer guaranteed.

COMMON TRAP
Never round a required sample size down or to the nearest integer. Round the final requirement upward. If sigma is unknown, use only the source-supported pilot/conservative estimate approach.

KNOWLEDGE CHECK - answer before looking at the key
A sample-size formula gives n=96.04. What sample size is required?

A. 97
B. 96
C. 96.04

YOUR TURN - work independently
Tutorial 8 Q5: σ=15, desired error ±5, 95% confidence. What required sample size should be used?
Answer: _____

EXIT CRITERION
You can derive/select the mean sample-size formula, explain each input and round up automatically.

Sources: Chapter 8 slides pp. 45-52 | Tutorial 8 Q5-Q6 | A1S2 2026 formula sheet sample-size formula for a mean

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.8 | OFFLINE GUIDED LEARN

8.8 - Required Sample Size for a Population Proportion

OBJECTIVE
Determine required n for estimating π, decide whether to use a pilot/prior estimate or the conservative 0.5 value, and round upward.

Recognise it

The target is a population proportion.
The question asks for required/minimum n at a stated confidence and margin e.
A prior/pilot proportion may be supplied; if none is supplied, the course uses 0.5 conservatively.

Explanation

The source formula is $n=z^2\pi(1-\pi)/e^2$. When π is unknown, a pilot estimate can be used.
If no prior estimate exists, the course uses π=.5 because .5(1-.5)=.25 is the largest possible Bernoulli variance term and therefore gives the conservative largest required sample size.
As with mean sample size, higher confidence or smaller margin of error requires a larger sample, and final n is rounded upward.

FORMULA / DECISION OBJECT
$n = z\alpha/2^2\ \pi(1-\pi)\ /\ e^2$; if no prior π is available, use 0.5; always round up.

Excel route - rehearse the syntax on paper

1. Use a pilot/prior proportion in the formula if the question supplies one.
2. If none is supplied, enter 0.5.
3. Required n: =ROUNDUP((z^2*p*(1-p))/e^2,0).

SAY ALOUD
For proportion sample size I need confidence, margin and a value for pi. If no estimate exists, I use 0.5 because it is conservative, then I round n up.

Worked example - A1S2 Practice Test Q5 style: 95% confidence, e=.06, no prior π

1. No prior estimate is available, so use π=.5.
2. 95% gives z=1.96.
3. $n=(1.96^2\times.5\times.5)/.06^2$.
4. $n\approx266.7778$.
5. Round upward.
Result: Required n=267.
Verify: Do not round 266.7778 to the nearest whole number; required n uses upward rounding.

COMMON TRAP
Use a supplied pilot/prior proportion when one is given; use 0.5 only when no estimate is available. Round the final required n upward.

KNOWLEDGE CHECK - answer before looking at the key
Why does the course use 0.5 when no prior proportion estimate is available?
A. Because p(1−p) is maximised at 0.5, giving the conservative largest n.
B. Because every population proportion equals 0.5.
C. Because 0.5 removes the confidence level from the formula.

YOUR TURN - work independently
Tutorial 8 Q7(a): prior π=.45, e=.02, 95% confidence. What required n should be used?
Answer: _____

EXIT CRITERION
You can decide pilot π versus 0.5, calculate n and explain why upward rounding is mandatory.

Sources: Chapter 8 slides pp. 53-57 | Tutorial 8 Q7-Q8 | A1S2 Practice Test Q5 | A1S2 2026 formula sheet sample-size formula for a proportion

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.9 | OFFLINE GUIDED LEARN

8.9 - Finite Population Correction for Confidence Intervals

OBJECTIVE
Recognise when sampling without replacement from finite N requires the course FPC rule, understand why the correction reduces standard error, and apply it to mean/proportion intervals.

Recognise it

Sampling is without replacement from a finite population of size N.

The sampling fraction n/N must be checked.

The source uses the FPC when n/N>0.05; below 5% its practical effect is little or none.

Explanation

The finite-population source says use the correction factor when more than 5% of the population is sampled without replacement.

The factor $\sqrt{[(N-n)/(N-1)]}$ multiplies the usual standard error. Because it is at most 1, it reduces the standard error and therefore narrows the confidence interval.

The same correction multiplies the mean standard error $s/\sqrt{n}$ and the proportion standard error $\sqrt{[p(1-p)/n]}$.

FORMULA / DECISION OBJECT
FPC = $\sqrt{[(N-n)/(N-1)]}$. Mean CI: $\bar{X} \pm t(s/\sqrt{n})FPC$. Proportion CI: $p \pm z\sqrt{[p(1-p)/n]}FPC$.

Excel route - rehearse the syntax on paper

1. FPC: =SQRT((N-n)/(N-1)).
2. Multiply the ordinary SE by the FPC before multiplying by the critical value.
3. Do not apply the correction merely because N is given; first check sampling without replacement and n/N.

SAY ALOUD
Finite population means without replacement. I check n over N. If it is more than five percent, the FPC multiplies and reduces the usual standard error.

Worked example - Finite-population source Example 8.10: N=200, n=27, $\bar{X}$=45.2222, s=23.1472, 95%

1. Sampling is without replacement and n/N=27/200=.135>0.05, so FPC matters.
2. df=26 and the source uses t=2.0555.
3. FPC=$\sqrt{[(200-27)/(200-1)]}$.
4. Apply $t\times(s/\sqrt{n})\times FPC$ as the margin.
5. The source obtains limits 36.6846 and 53.7598.
Result: 95% CI: (36.6846, 53.7598).
Verify: Because FPC<1, the interval is narrower than the corresponding no-FPC interval.

COMMON TRAP
Do not apply FPC merely because N appears in the question. The source requires sampling without replacement and uses n/N>0.05 as the practical trigger. Multiply the standard error by the FPC; do not add it.

KNOWLEDGE CHECK - answer before looking at the key
N=1000 and n=20, sampled without replacement. According to the supplied source, what should you conclude about FPC?

A. n/N=.02, so FPC has little or no practical effect.
B. FPC is mandatory whenever N is finite.
C. Use bootstrap instead of an interval.

YOUR TURN - work independently
Tutorial 8 Q11: $\bar{X}$=75, s=24, n=36, N=200, sampled without replacement. Construct the 95% confidence interval for μ.

EXIT CRITERION
You can recognise the FPC trigger, write the factor from memory and explain why it narrows the interval.

Sources: Chapter 8 slides pp. 59-63 | Dedicated finite-population extract §8.7 pp. 1-2 | Tutorial 8 Q11-Q12

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.10 | OFFLINE GUIDED LEARN

8.10 - Finite-Population Sample Size Determination

OBJECTIVE
Compute the initial infinite/large-population sample size n_0, adjust it for finite N, and round the resulting required sample size upward.

Recognise it

The question asks for required sample size and gives finite N.

Sampling is without replacement.

You first use the ordinary mean/proportion sample-size formula to obtain n_0.

Explanation

The finite-population source tells you to calculate n_0 without FPC first, using the ordinary mean or proportion sample-size formula.

Then apply $n=n_0N/[n_0+(N-1)]$. The adjusted finite-population sample size is smaller than n_0 because sampling without replacement extracts more information from a finite population.

After the adjustment, round upward to the required whole sample size.

FORMULA / DECISION OBJECT
$n = n_0N / [n_0 + (N-1)]$, where n_0 is the unadjusted required sample size.

Excel route - rehearse the syntax on paper

1. Calculate n0 first in its own cell.
2. Finite adjustment: =(n0*N)/(n0+(N-1)).
3. Final requirement: =ROUNDUP(adjusted_n,0).

SAY ALOUD
For finite-population sample size I first calculate n-zero normally, then shrink it with n-zero N over n-zero plus N minus one, and round up.

Worked example - Source finite-population sample-size example: n_0=96.04, N=5000

1. Start with the ordinary mean sample-size result n_0=96.04.
2. Apply n=(96.04×5000)/(96.04+4999).
3. Adjusted n≈94.24.
4. Round upward.
Result: Required finite-population sample size = 95.
Verify: The adjustment reduces the sample requirement from 97 (unadjusted) to 95.

COMMON TRAP
Keep n0 conceptually separate from the final finite-population n. Apply the finite adjustment to n0 and round the final adjusted requirement upward.

KNOWLEDGE CHECK - answer before looking at the key
Which formula is the supplied finite-population sample-size adjustment?

A. $n=n_0N/[n_0+(N-1)]$
B. $n=n_0/N$
C. $n=N-n_0$

YOUR TURN - work independently
Tutorial 8 Q12(b): N=1000, pilot p=.30, e=.05, 90% confidence. What finite-population required n should be used?
Answer: _____

EXIT CRITERION
You can distinguish n_0 from final n and perform the finite adjustment without mixing it into the original formula.

Sources: Chapter 8 slides p. 63 | Dedicated finite-population extract §8.7 pp. 2-3 | Tutorial 8 Q12 | A1S2 Practice Test Q6 | A1S2 2026 formula sheet finite-population sample-size formula

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.11 | OFFLINE GUIDED LEARN

8.11 - Bootstrapping: Why, Resampling Logic & Percentile Confidence Limits

OBJECTIVE
Explain when bootstrapping is appropriate, distinguish the original sample from bootstrap resamples, preserve resample size n, build the resampling distribution and obtain confidence limits using the course ordered-array method.

Recognise it

The population cannot reasonably be assumed normal.

You have an observed sample and can resample from it with replacement.

The task asks for a bootstrap confidence interval for a population mean.

Explanation

The supplied bootstrapping source says that when one cannot assume a normal population, bootstrap methods use repeated resampling with replacement from the initial sample and make no assumption about the underlying population distribution.

Select the original random sample without replacement from the population. Then each bootstrap resample selects n values with replacement from that observed sample, so values may repeat and others may be omitted.

Compute $\bar{X}^*$ for each resample, repeat many times, order all resampled means, then cut off α/2 from each tail to obtain the lower and upper confidence limits.

FORMULA / DECISION OBJECT
Original sample → resample n values WITH replacement → compute $\bar{X}^*$ → repeat m times → order $\bar{X}^*$ values → take the course-specified α/2 tail cutoffs.

Excel route - rehearse the syntax on paper

1. Software is required because many resamples are used.
2. Each resample mean can be calculated with AVERAGE(resample_range).
3. After all means are calculated, order them or use SMALL/LARGE according to the course cutoff convention for that m and confidence level.

SAY ALOUD
Bootstrap resampling happens from the observed sample, with replacement, and every resample stays size n. I calculate the statistic for each resample, order those statistics and cut alpha over two from each tail.

Worked example - Source 90% bootstrap example with 100 resample means

1. The confidence level is 90%, so α=.10 and α/2=.05.
2. There are 100 ordered resample means.
3. .05×100=5, so the source uses the 5th smallest and 5th largest values.
4. The 5th smallest is 36.6296 and the 5th largest is 51.5926.
Result: 90% bootstrap CI: (36.6296, 51.5926).
Verify: Do not use the minimum and maximum; the course removes α/2 from each tail.

COMMON TRAP
Bootstrap resampling is from the observed sample WITH replacement and keeps resample size n. Do not sort the original raw observations and call those the bootstrap confidence limits.

KNOWLEDGE CHECK - answer before looking at the key
Which description is a bootstrap resample in the supplied Chapter 8 method?

A. Select n values with replacement from the observed sample.
B. Select n new values without replacement from the population.
C. Generate n values from a fitted normal model.

YOUR TURN - work independently
An original sample contains a value only once. Can that value appear twice in one bootstrap resample?

A. Yes, because resampling is with replacement.
B. No, because a resample must preserve original frequencies exactly.
C. Only if n is doubled.

EXIT CRITERION
You can narrate the seven source steps in order and explain why duplicates/omissions are normal in a bootstrap resample.

Sources: Chapter 8 slides pp. 64-72 | Dedicated Bootstrapping §8.8 extract pp. 1-2 | A1S2 Practice Test Q1

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.12 | OFFLINE GUIDED LEARN

8.12 - Bootstrap in Excel & the A1S2 Workbook Workflow

OBJECTIVE
Translate the Chapter 8 bootstrap algorithm into the workbook workflow demonstrated by the 2026 A1S2 practice test without memorising a particular dataset or inventing an unsupported cutoff convention.

Recognise it

The EMSLearn question directs you to a named workbook sheet.

The original sample and an index/grid of resamples are already supplied.

You must calculate a statistic for each resample in Excel and then obtain the requested bootstrap limits.

Explanation

The 2026 practice test explicitly directs students to the Bootstrap worksheet of A1PTData.xlsx. It describes the original sample as drawn without replacement and an index of 250 resamples drawn from the original sample with replacement.

The course bootstrap source establishes the general method, but the supplied practice-test document does not state a unique 250-resample extraction-index rule. Therefore this lesson trains the supported workflow rather than inventing a specific index that is not stated in the supplied source.

Your execution sequence is: identify the original sample → build/read each with-replacement resample → calculate every resample mean → create/order the mean distribution → apply the course/in-question cutoff convention → enter the requested lower and upper limits in EMSLearn.

FORMULA / DECISION OBJECT
Workbook execution = AVERAGE(each resample) → ordered bootstrap means → course-specified tail cutoffs. Keep the statistical method separate from the EMSLearn four-decimal answer-entry rule.

Excel route - rehearse the syntax on paper

1. Use AVERAGE on each resample column/row to create the bootstrap mean for every resample.
2. Use sorting or SMALL/LARGE when the required order-statistic positions are known from the course/in-question convention.
3. The A1S2 practice-test answer-entry rule requires numerical answers to four decimals, with a full stop and no units/currency symbols.

SAY ALOUD
In the exam workflow I use Excel to execute the resampling calculation, but I still need to understand why it is with replacement, why n stays fixed, what distribution I am building, and which ordered tail values the course asks for.

Worked example - 2026 practice-test workflow (structure, not memorised answers)

1. Open the named Bootstrap worksheet.
2. Identify the original sample and the supplied with-replacement resample index/grid.
3. Calculate one mean for every resample in Excel.
4. Order the full set of resample means.
5. Apply the confidence-tail cutoff convention supported by the course/question.
6. Return to EMSLearn and enter the requested lower/upper values using the stated numerical format.
Result: The statistical object being ordered is the distribution of bootstrap sample means.
Verify: If you are sorting the raw original observations instead of the resample means, you are on the wrong object.

COMMON TRAP
Do not invent a 250-resample cutoff index that the supplied practice-test text never states. Execute the supported workbook workflow and use the course/question-specific cutoff convention actually provided.

KNOWLEDGE CHECK - answer before looking at the key
What should you calculate for each bootstrap resample when the target is a confidence interval for the population mean?

A. The resample mean X^*
B. Only the resample minimum
C. A new population standard deviation σ

YOUR TURN - work independently
Which Excel step belongs immediately after you have generated/read all bootstrap resamples for a mean CI?

A. Calculate AVERAGE for each resample.
B. Calculate one grand average of every cell and stop.
C. Replace repeated values so every resample is unique.

EXIT CRITERION
You can move between EMSLearn instructions and Excel without losing the statistical object, and you can state which parts are source-confirmed versus convention-dependent.

Sources: A1S2 2026 Practice Test Q1 and first-page answer-entry rules | Chapter 8 slides pp. 64-72 | Dedicated Bootstrapping §8.8 extract | Existing A1S2 Excel execution companion is a training analogue only, not an official source

WORKSPACE / RETRIEVAL NOTES

CHAPTER 8 | LESSON 8.13 | OFFLINE GUIDED LEARN

8.13 - Responsible Confidence-Interval Reporting & Chapter Synthesis

OBJECTIVE
Report a confidence interval in the complete form required by the Chapter 8 source and connect the whole chapter into one decision sequence.

Recognise it

The task asks you to report or interpret an estimate, not merely calculate two limits.

You need to state what belongs with a point estimate in a responsible report.

You need to choose among mean/proportion, known/unknown sigma, sample-size, finite-population and bootstrap routes.

Explanation

The Chapter 8 Ethical Issues slide says that when reporting a point estimate, a confidence interval reflecting sampling error should be included; the confidence level, sample size and an interpretation should also be reported.

A complete Chapter 8 solution therefore identifies the population parameter, gives the interval and confidence level, states n where relevant, and interprets the interval in context rather than reporting naked numbers.

For method selection, first identify the parameter and task. Mean CI: known population sigma gives Z; unknown sigma gives Student t. Proportion CI uses p and its estimated SE. Required-n questions use the corresponding sample-size formula and round up. Sampling without replacement from a finite population may require FPC. Non-normal settings may trigger the supplied bootstrap route.

FORMULA / DECISION OBJECT
Parameter -> task -> sigma/source conditions -> critical distribution / sample-size route -> finite/bootstrap branch -> calculate -> report interval + confidence level + n + contextual interpretation.

Excel route - rehearse the syntax on paper

1. Use Excel to execute the selected calculation, but do not let the worksheet replace method selection.
2. Before submitting, verify the object, confidence level, rounding rule and whether the question asks for limits, margin, n or an interpretation.
3. Say aloud: "A complete confidence-interval report gives the parameter in context, the interval, the confidence level, the sample size where relevant, and an interpretation. I choose the statistical route before I type an Excel formula."

SAY ALOUD
A complete confidence-interval report gives the parameter in context, the interval, the confidence level, the sample size where relevant, and an interpretation. I choose the statistical route before I type an Excel formula.

Worked example - Complete report

1. Suppose a 95% CI for a population mean is calculated as (83.04, 86.96) from n=64.
2. Name the target: the population mean.
3. State the confidence level: 95%.
4. Include the sample size: n=64.
5. Interpret in context: we are 95% confident that the population mean lies between 83.04 and 86.96 under this interval procedure.

Result: The calculation and the report are both complete.
Verify: Do not replace a contextual interpretation with naked numerical limits.

COMMON TRAP
A correct pair of limits is not a complete statistical report when the question asks for interpretation or reporting.

EXIT CRITERION
You can choose the Chapter 8 method and produce a complete contextual report without relying on a formula label.

Sources: Chapter 8 slides p. 58 (Ethical Issues) | Chapter 8 summary p. 72 | Tutorial 8 interpretation/reporting questions

WORKSPACE / RETRIEVAL NOTES

OFFICIAL STUDENT-t TABLE

Use the exact supplied table when a paper/table critical value is required. In Excel, use the T.* functions shown in the relevant lessons.

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

t-table.xls 7/14/2007

CHAPTER 9

Hypothesis Testing - Slides 1-51 Only

SCOPE
A1S2 SCOPE LOCK: only material through official Slide 51. This pack stops after Lesson 9.7.

Offline tracker: [ ] 9.1 [ ] 9.2 [ ] 9.3 [ ] 9.4 [ ] 9.5 [ ] 9.6 [ ] 9.7

STOP AFTER 9.7 FOR A1S2 CHAPTER 9 STUDY

CHAPTER 9 | LESSON 9.1 | OFFLINE GUIDED LEARN

9.1 - Hypothesis Foundations

CH9 SCOPE
IN SCOPE: official Chapter 9 Slides 1-51. Later Chapter 9 sections are intentionally not printed in this exam-study pack.

OBJECTIVE
Build H0 and H1 around a population parameter, start from the H0 assumption, and use the source-approved decision language.

Explanation

A hypothesis is a claim about a population parameter such as the population mean μ or population proportion π. The sample statistic is evidence; it is not what H0 is written about.

H0 represents the claim/status quo to be validated and carries equality in the SDS188 treatment. H1 is mutually exclusive and challenges H0.

Testing begins by assuming H0 is true - the slides use the "innocent until proven guilty" analogy - and then asks whether the sample evidence is sufficiently inconsistent with that assumption.

The source explicitly warns that H0 is never accepted. Your statistical decision is "reject H0" or "do not reject H0".

SAY ALOUD
My hypotheses are about a population parameter. I assume H0 is true, use the sample as evidence, and I either reject H0 or do not reject H0 - I never accept H0.

Worked example - Turn a claim into hypotheses

1. Claim: the population mean fill is 500 ml.
2. Parameter is μ, not $\bar{X}$.
3. Write H0: μ=500.
4. If the research question asks whether the mean differs, write H1: μ≠500.
5. Begin analysis under the assumption H0 is true.
Result: H0: μ=500; H1: μ≠500.
Verify: Both statements concern μ and are mutually exclusive.

KNOWLEDGE CHECK - answer before looking at the key
Which conclusion is valid source language?

A. Do not reject H0
B. Accept H0
C. Prove H0 true

YOUR TURN - work independently
A company claims the population mean trip time is 110 minutes. Which H0 is correctly written?

A. H0: μ=110
B. H0: $\bar{X}$=110
C. H0: μ≠110

EXIT CRITERION
Move on when you can write H0/H1 from wording and automatically say "reject" or "do not reject," never "accept."

Sources: Official Chapter 9 deck pp. 4-10 | SDS Tutorial 9 Q1-3

WORKSPACE / RETRIEVAL NOTES

CHAPTER 9 | LESSON 9.2 | OFFLINE GUIDED LEARN

9.2 - Decision Risk: α, β & Power

CH9 SCOPE
IN SCOPE: official Chapter 9 Slides 1-51. Later Chapter 9 sections are intentionally not printed in this exam-study pack.

OBJECTIVE
Understand the two possible testing errors and predict how α, β, power, effect size, variance and n move together.

Explanation

Type I error means rejecting H0 when H0 is true. Its probability is α, the significance level.

Type II error means not rejecting H0 when H0 is false. Its probability is β. Power is 1-β: the probability of rejecting a false H0.

The confidence coefficient is 1-α. Type I and Type II errors cannot occur in the same test outcome because H0 cannot simultaneously be true and false.

For fixed n, reducing α makes rejection harder and β increases. Power rises when the true parameter is farther from the null value or n increases; larger variance makes detection harder and increases β.

SAY ALOUD
Alpha is the chance I reject a true H0. Beta is the chance I fail to reject a false H0. Power is one minus beta. With fixed n, lowering alpha raises beta.

Worked example - Why distance from μ0 changes power

1. Suppose H0: μ=100.
2. A true μ=90 is 10 units from the null.
3. A true μ=75 is 25 units from the null.
4. The farther alternative creates more separation between the sampling distributions.
5. Therefore β is smaller and power larger at μ=75 than at μ=90.
Result: Closer alternatives are harder to detect; farther alternatives have higher power.
Verify: This is exactly the logic behind Tutorial 9 Q3.

KNOWLEDGE CHECK - answer before looking at the key
For fixed n, if α falls from .05 to .01, what happens to β?

A. β increases
B. β decreases
C. β becomes α

YOUR TURN - work independently
Which change, all else fixed, increases power?

A. Increase n
B. Increase variance
C. Move the true mean closer to μ0

EXIT CRITERION
Move on when you can draw the 2×2 truth/decision table from memory and predict the direction of β and power changes.

Sources: Official Chapter 9 deck: Type I/II error and power sections | SDS Tutorial 9 Q1-3, Q21-22

WORKSPACE / RETRIEVAL NOTES

CHAPTER 9 | LESSON 9.3 | OFFLINE GUIDED LEARN

9.3 - Tails, α & Rejection Regions

CH9 SCOPE
IN SCOPE: official Chapter 9 Slides 1-51. Later Chapter 9 sections are intentionally not printed in this exam-study pack.

OBJECTIVE
Let H1 determine the rejection tail and place α correctly before looking up a critical value.

Explanation

The direction is determined by H1, not by the sign of the sample result.

H1: parameter≠value gives a two-tail test. H1: parameter<value gives a lower-tail test. H1: parameter>value gives an upper-tail test.

For a two-tail test, divide α equally: α/2 in each tail. For a one-tail test, place the full α in the one rejection tail.

Always sketch or state the rejection region before calculating the test statistic.

SAY ALOUD
Not equal means two tails. Less than means lower tail. Greater than means upper tail. Two tails split alpha; one tail gets all alpha.

Worked example - Tutorial 9 Q4 critical region

1. Two-tail test and α=.10.
2. Split α: .05 per tail.
3. The standard-normal cutoffs are approximately ±1.645.
4. Reject H0 if Z<-1.645 or Z>1.645.
Result: Rejection region: Z<-1.645 or Z>1.645.
Verify: The middle non-rejection area is 0.90.

KNOWLEDGE CHECK - answer before looking at the key

H1: μ>500 means which test?

A. Upper-tail
B. Lower-tail
C. Two-tail

YOUR TURN - work independently

For a lower-tail test at α=.01, where is the rejection region?

A. Entire α=.01 in the left tail
B. α/2 in both tails
C. Entire α=.01 in the right tail

EXIT CRITERION
Move on when you can see H1 and immediately name the tail, allocate α and describe the rejection region.

Sources: Official Chapter 9 deck: one-tail versus two-tail slides | SDS Tutorial 9 Q4-6

WORKSPACE / RETRIEVAL NOTES

CHAPTER 9 | LESSON 9.4 | OFFLINE GUIDED LEARN

9.4 - Mean Test - σ Known: Z Critical-Value

CH9 SCOPE
IN SCOPE: official Chapter 9 Slides 1-51. Later Chapter 9 sections are intentionally not printed in this exam-study pack.

OBJECTIVE
Recognise a known-σ one-sample mean test, compute Z, and make the critical-value decision.

Explanation

When the question tests a population mean μ and the population standard deviation σ is known, the Chapter 9 route is a Z test.
Under H0, the sample mean is compared with μ0 using its standard error $\sigma/\sqrt{n}$. The test statistic is $Z=(\bar{X}-\mu_0)/(\sigma/\sqrt{n})$.
The critical value(s) come from α and the tail already chosen from H1. If Zcalc lands in the rejection region, reject H0; otherwise do not reject H0.
Finish with a sentence about the population claim - not just "reject".

SAY ALOUD
Mean plus known population sigma means Z. I use x-bar minus mu-zero over sigma divided by square root n, then compare with the rejection region set by H1 and alpha.

Worked example - Tutorial 9 paint cans

1. H0: μ=3.75; H1: μ≠3.75; α=.01.
2. σ=.08, n=50, $\bar{X}$=3.73.
3. SE=.08/√50≈.011314.
4. Z=(3.73-3.75)/.011314≈-1.768.
5. Two-tail α=.01 gives critical values about ±2.576.
6. -1.768 is not in either rejection tail.
Result: Do not reject H0 at the 1% significance level.
Verify: The test statistic is not extreme enough for the strict .01 two-tail rejection region.

KNOWLEDGE CHECK - answer before looking at the key
A mean is tested and σ is known. Which reference test does the source select?

A. Z test
B. Student t test
C. Proportion Z test

YOUR TURN - work independently
ATM example: μ0=500, σ=120, n=36, $\bar{X}$=550. What is Zcalc?
Answer: _____

EXIT CRITERION
Move on when you can identify known σ → Z and execute hypotheses → tail → SE → Z → critical comparison → contextual conclusion.

Sources: Official Chapter 9 deck: Z-test critical-value workflow | SDS Tutorial 9 Q10-11

WORKSPACE / RETRIEVAL NOTES

CHAPTER 9 | LESSON 9.5 | OFFLINE GUIDED LEARN

9.5 - Z Test - p-Value + Excel

CH9 SCOPE
IN SCOPE: official Chapter 9 Slides 1-51. Later Chapter 9 sections are intentionally not printed in this exam-study pack.

OBJECTIVE
Understand what the p-value means, calculate it for the correct tail, compare it with α, and reconstruct the deck's Excel route.

Explanation

The deck defines the p-value as the probability, assuming H0 is true, of obtaining a test statistic equal to or more extreme than the observed one.

Decision rule: if $p<\alpha$, reject H0; if $p\geq\alpha$, do not reject H0.

Two-tail Z p-values use both tails. The source Excel formula is (1-NORM.S.DIST(ABS(Zcalc),TRUE))*2. A directed one-tail p-value uses only the tail specified by H1.

A p-value is not the probability that H0 is true. It is calculated under the assumption that H0 is true.

SAY ALOUD
The p-value is an extremeness probability under H0. If p is smaller than alpha I reject H0; otherwise I do not reject H0.

Worked example - Tutorial 9 Q7

1. Two-tail test with Z=2.00.
2. One upper tail beyond 2.00 is about .02275.
3. Two-tail p≈2(.02275)=.0455.
4. Compare .0455 with α=.05.
5. .0455<.05, so reject H0.
Result: p≈.0455; reject H0 at α=.05.
Verify: Two-tail means count equally extreme results in both directions.

KNOWLEDGE CHECK - answer before looking at the key
If p=.032 and α=.05, what is the decision?

A. Reject H0
B. Do not reject H0
C. Accept H0

YOUR TURN - work independently
Lower-tail test with Z=−1.38. Enter the p-value to 4 d.p.
Answer: _____

EXIT CRITERION
Move on when you can explain the p-value in words, choose one-tail versus two-tail area, and reconstruct the source Excel formula without copying it.

Sources: Official Chapter 9 deck: p-value definition and Excel Z workflow | SDS Tutorial 9 Q7-10

WORKSPACE / RETRIEVAL NOTES

CHAPTER 9 | LESSON 9.6 | OFFLINE GUIDED LEARN

9.6 - Two-Tail Z Test ↔ Confidence Interval

CH9 SCOPE
IN SCOPE: official Chapter 9 Slides 1-51. Later Chapter 9 sections are intentionally not printed in this exam-study pack.

OBJECTIVE
Use the source's equivalence between a two-tail Z test and the same-level confidence interval as a conceptual cross-check.

Explanation

The Chapter 9 deck explicitly links a two-tail hypothesis test with the confidence interval built at confidence level 1-α.

If the hypothesised μ0 lies outside that interval, the two-tail test rejects H0 at significance α. If μ0 lies inside the interval, the test does not reject H0.

This is a cross-check of the same inferential evidence; it does not turn the hypothesis test into a Chapter 8 problem.

Tutorial 9 Q10 deliberately asks you to compare the 1% two-tail test with a 99% confidence interval.

SAY ALOUD
For a two-tail Z test, the same-level confidence interval tells the same decision story: null value outside means reject; null value inside means do not reject.

Worked example - Paint-can cross-check

1. The 99% confidence interval is approximately 3.7009 to 3.7591 kg.
2. The null value μ0=3.75 kg lies inside the interval.
3. Therefore the equivalent two-tail α=.01 test does not reject H0.
4. This agrees with Z≈-1.768 being inside ±2.576.
Result: The CI and critical-value test agree: do not reject H0.
Verify: Same α/confidence pairing: 1-α=.99.

KNOWLEDGE CHECK - answer before looking at the key

A 95% CI for μ is (48,52) and H0:μ=50 in a two-tail α=.05 test. What should the test do?

A. Do not reject H0
B. Reject H0
C. Accept H0

YOUR TURN - work independently

A same-level two-tail CI excludes μ0. What is the matching test decision?

A. Reject H0
B. Do not reject H0
C. The two methods must disagree

EXIT CRITERION
Move on when you can predict the test decision just by locating μ0 relative to the same-level two-tail interval.

Sources: Official Chapter 9 deck: hypothesis-test / confidence-interval connection | SDS Tutorial 9 Q10(c)-(d)

WORKSPACE / RETRIEVAL NOTES

CHAPTER 9 | LESSON 9.7 | OFFLINE GUIDED LEARN

9.7 - Mean Test - σ Unknown: Student t

CH9 SCOPE
IN SCOPE: official Chapter 9 Slides 1-51. Later Chapter 9 sections are intentionally not printed in this exam-study pack.

OBJECTIVE
Switch correctly from Z to Student t when σ is unknown, use s and df=n-1, check the source assumptions, and choose the correct t tail/Excel function.

Explanation

If the population standard deviation σ is unknown and sample standard deviation s is supplied, the Chapter 9 mean-test route uses Student t.

$T=(\bar{X}-\mu_0)/(s/\sqrt{n})$ and df=n-1. The t distribution accounts for uncertainty from estimating σ with s.

The source states the normal-population assumption and discusses checking normality; it also notes t can be used when the sample is not very small and the population is not very skewed.

For two tails the deck uses T.DIST.2T(ABS(tcalc),df); for critical values it shows T.INV with the appropriate cumulative probability.

SAY ALOUD
Mean plus unknown population sigma means Student t. I use sample s, divide by square root n, and use degrees of freedom n minus one.

Worked example - Tutorial 9 textbook-cost example

1. H0: μ=1500; H1: μ>1500; α=.10.
2. $\bar{X}$=1577, s=216, n=25, so df=24.
3. SE=216/5=43.2.
4. T=(1577-1500)/43.2≈1.7824.
5. Upper-tail critical t at α=.10 and df=24 is about 1.318.
6. 1.7824>1.318, so reject H0. The one-tail p-value is about .0437.
Result: There is evidence at α=.10 that the population mean textbook cost exceeds R1500.
Verify: Unknown σ is why s and t appear.

KNOWLEDGE CHECK - answer before looking at the key
n=25 and σ is unknown. What df is used?

A. 24
B. 25
C. 26

YOUR TURN - work independently
Battery example: $\bar{X}$≈473.4615, s≈210.7663, n=13, μ0=500. What is Tcalc?
Answer: _____

EXIT CRITERION
Move on when the phrase "σ unknown, s supplied" automatically triggers t, df=n-1 and the correct t-tail workflow.

Sources: Official Chapter 9 deck: Student-t sections and Excel workflow | SDS Tutorial 9 Q12-14

WORKSPACE / RETRIEVAL NOTES

CHAPTER 9 A1S2 STOP POINT

STOP HERE

You have reached the A1S2 Chapter 9 cutoff. The lecturer announcement limits Chapter 9 to Slides 1-51. Do not spend scarce A1S2 revision time on the later detailed one-tail worked section, one-proportion tests, later power calculations, later planning/practical-significance or later ethics material unless the lecturer issues a newer clarification.

WHAT MUST STILL BE AUTOMATIC

H0/H1 -> tail -> sigma known? Z : t -> critical-value or p-value decision -> reject/do not reject H0 -> context. Also know Type I, Type II, alpha, beta and the basic meaning power=1-beta.

CONSOLIDATED ANSWER KEY

Use this only after you have answered the lesson checks and Your Turn items without support. Mark the error type, repair the nearest competing method, then retry later.

Chapter 6 answers

6.1 - Knowledge Check: P(X=4)=0 | *An exact point has no width, so it contributes zero area.*

6.1 - Your Turn: P(X≤8) | *Because P(X=8)=0, adding that single point changes no area.*

6.2 - Knowledge Check: 5 | *The notation gives variance σ²=25, so σ=√25=5.*

6.2 - Your Turn: -2 | *z=(42-50)/4=-2. The negative sign makes sense because 42 is below the mean 50.*

6.3 - Knowledge Check: 1-Φ(z) | *The cumulative function gives left area; the right tail is its complement.*

6.3 - Your Turn: 0.4522 | *Right tail = 1-0.5478=0.4522.*

6.4 - Knowledge Check: 0.90 | *If 10% is above x, then 90% is below x.*

6.4 - Your Turn: 86.16 | *Only 5% above means cumulative-left area 0.95. z≈1.645, so x=73+1.645(8)=86.16.*

6.5 - Knowledge Check: Use several numerical and graphical indicators together | *The course notes and transcript require an integrated conclusion from multiple indicators.*

6.5 - Your Turn: Do not call it normal; the evidence conflicts | *One matching centre measure cannot override clear graphical non-normality.*

6.6 - Knowledge Check: j/(n+1) | *That is the explicit current course workflow and the supplied Excel solution uses it.*

6.6 - Your Turn: =NORM.S.INV(D3) | *α is a cumulative probability, so you need the inverse standard-normal function.*

6.7 - Knowledge Check: Density height | *The probability requires area, so it also depends on interval width.*

6.7 - Your Turn: 0.1 | *The target width is 1 out of a total width 10, so probability=1/10=0.10.*

6.8 - Knowledge Check: Exponential | *The random object is waiting time, which is continuous.*

6.8 - Your Turn: 0.4512 | *Convert λ=12/hour to 0.2/minute. P(X<3)=1-e^(-0.2×3)=1-e^(-0.6)≈0.4512.*

6.9 - Knowledge Check: -ln(1-α)/λ | *It comes directly from solving 1-e^(-λx)=α for x.*

6.9 - Your Turn: 0.1436 | *Right tail gives e^(-3x)=0.65. Thus x=-ln(0.65)/3≈0.1436.*

6.10 - Knowledge Check: P(Xa>24.5) | *The integer 25 owns the continuous slice from 24.5 to 25.5, so "25 or more" starts at 24.5.*

6.10 - Your Turn: P(Xa<24.5) | *X<25 means X≤24 in the discrete model, so the corrected boundary is 24.5.*

Chapter 7 answers

7.1 - Knowledge Check: The distribution of all possible sample means for samples of size 25 | *The source defines a sampling distribution as all possible values of a sample statistic for a fixed sample size.*

7.1 - Your Turn: Sampling distribution of p | *The repeated statistic is the sample proportion p.*

7.2 - Knowledge Check: Unbiasedness | *The source explicitly identifies $\bar{X}$ as unbiased when its sampling-distribution mean equals μ.*

7.2 - Your Turn: 6.0 | *The six values sum to 36, so μ=36/6=6. Exact enumeration confirms the possible sample means average to 6.*

7.3 - Knowledge Check: It is halved | *SE=σ/√n, so √4=2.*

7.3 - Your Turn: 2.0 | *12/√36=12/6=2.*

7.4 - Knowledge Check: 6/√50 | *Variance 36 means σ=6, and a class average is $\bar{X}$.*

7.4 - Your Turn: $\bar{X}$ and σ/√36 | *The word average identifies $\bar{X}$.*

7.5 - Knowledge Check: 1-Φ(z) | *Above is a right-tail event while the cumulative normal gives left area.*

7.5 - Your Turn: 0.1356660609 | *SE=2, z=1.1, so 1-Φ(1.1)≈0.1357.*

7.6 - Knowledge Check: 0.35 | *The areas must sum to 1.*

7.6 - Your Turn: [362.12, 373.88] | *SE=3; z*=1.96; 368±1.96(3)=368±5.88 → 362.12,373.88.*

7.7 - Knowledge Check: The sampling distribution of $\bar{X}$ | *That is the Chapter 7 source statement.*

7.7 - Your Turn: 0.0339445774 | *μ=σ=9 sec for exponential; SE=9/√30≈1.6432; z≈1.8257; right-tail≈0.0339.*

7.8 - Knowledge Check: Student t with df=24 | *The current Chapter 7 source patch explicitly verifies the Student-t route for unknown σ.*

7.8 - Your Turn: 2.0 | *T=(54-50)/(10/√25)=4/2=2; df=24.*

7.9 - Knowledge Check: 0.75 | *p=48/64=0.75; π=0.70 is the population parameter.*

7.9 - Your Turn: [0.35, 0.0754155156] | *p=14/40=.35; estimated SE=√(.35×.65/40)≈.0754.*

7.10 - Knowledge Check: It narrows around the same centre π | *SE is inversely proportional to √n while E(p)=π.*

7.10 - Your Turn: 0.7887004527 | *SE=.04; z limits ±1.25; probability Φ(1.25)-Φ(-1.25)≈.7887.*

7.11 - Knowledge Check: $\bar{X}_1-\bar{X}_2>25$ | *Preserve the declared order 1-2 and use the stated difference threshold.*

7.11 - Your Turn: 0.1122470397 | *Centre=10; SE≈12.3491; z=(25-10)/12.3491≈1.2147; right-tail≈0.1122.*

7.12 - Knowledge Check: Yes, because n/N=0.08>0.05 | *The Chapter 7 online section uses the >5% sampling-fraction trigger.*

7.12 - Your Turn: 0.949157995752499 | *√(900/999)≈0.9492.*

7.13 - Knowledge Check: $\bar{X}$ → Student t | *The object is a sample mean and σ status selects the current t route.*

7.13 - Your Turn: Identify object → choose model/SE → translate region → execute in Excel → interpret | *This sequence preserves statistical method selection while using Excel as the execution environment.*

Chapter 8 answers

8.1 - Knowledge Check: Estimator A | *The Chapter 8 source defines the unbiased estimator with the smaller variance as relatively more efficient.*

8.1 - Your Turn: The estimator tends to get closer to the parameter as n increases. | *The Chapter 8 slides define consistency by the estimator-parameter difference becoming smaller as sample size increases.*

8.2 - Knowledge Check: We are 95% confident using a procedure that captures the parameter in 95% of such intervals. | *The source explicitly gives the long-run relative-frequency interpretation and says a specific interval either contains the parameter or does not.*

8.2 - Your Turn: 0.01 | *1-α=0.98 gives α=0.02, so α/2=0.01 in each tail.*

8.3 - Knowledge Check: Standard normal Z | *The Chapter 8 known-σ interval uses zα/2.*

8.3 - Your Turn: [114.696, 135.304] | *99% uses zα/2=2.576. SE=24/√36=4. Margin=2.576×4=10.304. CI=(114.696,135.304).*

8.4 - Knowledge Check: df=9 and one-tail 0.025 | *df=n-1=9, and 95% confidence gives α=.05, so α/2=.025 in each tail.*

8.4 - Your Turn: 1.753 | *df=15. For 90% confidence, α=.10 so α/2=.05. At df=15 and one-tail .05, t=1.753.*

8.5 - Knowledge Check: Because replacing σ with variable sample s adds uncertainty. | *That is the explanation given on the Chapter 8 slides.*

8.5 - Your Turn: [134.618, 147.982] | *df=23 and 90% two-sided gives t≈1.714. SE=19.1/√24≈3.8988. Margin≈6.6820. CI≈(134.6180,147.9820).*

8.6 - Knowledge Check: Sample proportion p | *The Chapter 8 interval-estimation source estimates the standard error using sample p.*

8.6 - Your Turn: [0.90165, 0.95725] | *p=303/326≈0.929448. SE≈0.014183. Margin≈1.96×SE≈0.027798. CI≈(0.901650,0.957246).*

8.7 - Knowledge Check: 97 | *The source explicitly says always round required sample size upward.*

8.7 - Your Turn: 35 | *n=(1.96×15/5)²=34.5744, so round upward to n=35.*

8.8 - Knowledge Check: Because p(1-p) is maximised at 0.5, giving the conservative largest n. | *This is the conservative rule stated in the Chapter 8 material and confirmed in the 2026 practice-test style.*

8.8 - Your Turn: 2377 | *n=(1.96²×.45×.55)/.02²=2376.99, so required n=2377.*

8.9 - Knowledge Check: n/N=.02, so FPC has little or no practical effect. | *The finite-population source uses 5% as the practical threshold.*

8.9 - Your Turn: [67.6282, 82.3718] | *n/N=.18>0.05. df=35, t≈2.0301. FPC=√(164/199)≈0.90781. Adjusted SE=(24/6)×FPC≈3.63124. Margin≈7.37182. CI≈(67.6282,82.3718).*

8.10 - Knowledge Check: $n=n_0N/[n_0+(N-1)]$ | *This is Equation 8.14 in the supplied finite-population source and appears on the A1S2 formula sheet.*

8.10 - Your Turn: 186 | *Use z=1.645. $n_0=(1.645^2\times.30\times.70)/.05^2\approx227.3061$. Adjust: n=(227.3061×1000)/(227.3061+999)≈185.3584. Round up to 186.*

8.11 - Knowledge Check: Select n values with replacement from the observed sample. | *That with-replacement resampling step is the defining procedure in the supplied source.*

8.11 - Your Turn: Yes, because resampling is with replacement. | *The Chapter 8 source explicitly shows a resample where 73 appears twice even though it appeared once in the original sample.*

8.12 - Knowledge Check: The resample mean X* | *The Chapter 8 bootstrap algorithm says compute $\bar{X}$ for each resample and construct the resampling distribution of those means.*

8.12 - Your Turn: Calculate AVERAGE for each resample. | *The source algorithm requires one $\bar{X}$ from every resample before the resampling distribution can be ordered.*

Chapter 9 answers

9.1 - Knowledge Check: Do not reject H0 | *The Chapter 9 deck explicitly states that H0 is never accepted.*

9.1 - Your Turn: H0: μ=110 | *H0 is written about the population parameter μ and includes equality.*

9.2 - Knowledge Check: β increases | *A stricter rejection region lowers Type I risk but raises Type II risk for a fixed design.*

9.2 - Your Turn: Increase n | *A larger n reduces the sampling spread, so a false H0 is easier to detect.*

9.3 - Knowledge Check: Upper-tail | *The alternative points to values greater than the null value.*

9.3 - Your Turn: Entire α=.01 in the left tail | *One-tail means no split; "less than" points left.*

9.4 - Knowledge Check: Z test | *The deck's technique-selection branch uses Z when population σ is known.*

9.4 - Your Turn: 2.5 | *SE=120/6=20; Z=(550-500)/20=2.5.*

9.5 - Knowledge Check: Reject H0 | *p<α.*

9.5 - Your Turn: 0.0838 | *For a lower-tail test, p=Φ(-1.38)≈.0838; there is no doubling.*

9.6 - Knowledge Check: Do not reject H0 | *The hypothesised value 50 lies inside the 95% interval.*

9.6 - Your Turn: Reject H0 | *Exclusion of μ0 corresponds to the test statistic falling in a rejection tail.*

9.7 - Knowledge Check: 24 | *df=n-1.*

9.7 - Your Turn: -0.45399 | *T=(473.4615-500)/(210.7663/√13)≈-0.454.*

FINAL HIGH-RISK TRAPS - READ THE NIGHT BEFORE

Name the random object first: X is not Xbar; Xbar uses a standard error.
Population parameter versus sample statistic: mu vs Xbar; pi vs p; sigma vs s.
Known population sigma -> Z. Unknown sigma + sample s -> Student t, df=n-1.
Excel cumulative functions give left area. Convert right-tail wording before using an inverse/CDF.
CLT changes the sampling distribution of Xbar, not the original population.
For required sample size, ALWAYS ROUND UP.
FPC: do not apply merely because N appears. Check finite population, without replacement, and the course sampling-fraction rule.
Bootstrap resamples FROM THE OBSERVED SAMPLE, WITH replacement, and keeps resample size n.
Confidence interval for a mean is about the population mean, not individual values.
CI for a proportion uses observed sample p in the estimated SE.
In a hypothesis test, H0/H1 are about the population parameter; equality is in H0 under the SDS188 convention.
Tail comes from H1. Two-tail splits alpha/2; one-tail places alpha in the directed tail.
Decision language: reject H0 or do not reject H0. Never say accept H0.
Type I = reject a true H0. Type II = do not reject a false H0. Power = 1-beta.
For Chapter 9 A1S2, STOP at Slide 51 / Lesson 9.7 in this pack.

FRIDAY / SATURDAY CHECKLIST

[ ] Chapter 8 AF prepared for before opening the live assessment.
[ ] Chapter 8 AF submitted by Friday 28 Aug 17:30 and submission confirmed.
[ ] A1S2 Practice Test completed by Friday 28 Aug 23:59.
[ ] Chapter 6 revision AF used as a diagnostic; weak routes repaired.
[ ] Chapter 7 revision AF used as a diagnostic; weak routes repaired.
[ ] Chapter 9 Lessons 9.1-9.7 can be routed without labels.
[ ] Formula sheet scanned once: know what it gives and what it does not.
[ ] Excel syntax rehearsed from memory for the functions on the quick-reference page.
[ ] At least one closed-book mixed retrieval block completed before Saturday.
[ ] Assessment session/venue checked before leaving.

SCRATCH / WORKING PAGE 1

SCRATCH / WORKING PAGE 2

SCRATCH / WORKING PAGE 3

SCRATCH / WORKING PAGE 4